

# Proof-of-Work versus Proof-of-Stake: A Comparative Economic Analysis

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We develop an economic model to compare equilibrium security of Proof-of-Work (PoW) versus Proof-of-Stake (PoS) blockchains. We derive general conditions to determine when PoW blockchains are more secure than otherwise equivalent PoS blockchains and vice versa. Applying real-world parameter values to these conditions, we demonstrate that PoS blockchains are more secure than otherwise equivalent PoW blockchains. Furthermore, we demonstrate that PoS's security advantage over PoW is particularly salient for high-scale blockchains. (*JEL G0, O3*)

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A blockchain is a digital ledger that allows users to settle transactions (and perform other economic activities) without relying on a centralized entity. A blockchain can operate in this decentralized fashion because it employs a protocol that incentivizes agents to compete at each point in time to be selected for updating the blockchain, thereby negating the need for a centralized entity to provide such updates. Notably, the rules for the blockchain protocol's selection algorithm are publicly known and the ability to compete for selection by the algorithm is open to all that wish to participate. As a consequence, if a malicious agent amasses too much power in the protocol's competition by amassing a sufficiently large selection probability, then they

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can maliciously disrupt the blockchain. This feature generates a key security problem for blockchain users that has fundamental implications for blockchain adoption because blockchain users rationally internalize the negative effects of disruption. In turn, blockchain adoption will be commensurately low when the likelihood of disruption is high. Nonetheless, an agent increasing their probability of being selected by the blockchain protocol entails a cost and this cost endogenously depends on the specific blockchain protocol and the underlying economic environment. Crucially, keeping the economic environment fixed, different blockchain protocols generate different economic costs of disruption, different incentives to disrupt, and therefore different blockchain security levels. Although the prior literature acknowledges the importance of a blockchain's protocol in generating security and adoption, the prior literature has nonetheless not provided a unified framework to study equilibrium security and adoption comparatively across blockchain protocols. This paper provides the first such study.

In more detail, within this paper, we provide the first comparative analysis across the two most commonly utilized blockchain protocols, Proof-of-Work (PoW) and Proof-of-Stake (PoS). Notably, our work clarifies how the blockchain protocol (i.e., PoW or PoS) affects not only the equilibrium relationship between security and adoption but also how this relationship is affected by the blockchain's scale. Our key finding is that, given current real-world conditions, a PoS blockchain generates higher equilibrium security than an otherwise equivalent PoW blockchain and that this advantage is particularly salient at high-scale.

We establish our results theoretically via an economic model centered on a single blockchain, which is either of type PoW or of type PoS. We consider a model whereby agents possess overlapping investment horizons and choose to invest their wealth into blockchain activities or into an outside option that generates an exogenous return. Agents investing in blockchain activities, hereafter blockchain *users*, do so by allocating their wealth across two separate blockchain investment opportunities: blockchain validation and a DeFi application. Investment in blockchain validation entails participating in the block creation process through which *blocks* of transactions are added to the blockchain in exchange for an endogenous compensation for that service. In contrast, investment in the DeFi application corresponds to, for example, supplying liquidity to a liquidity pool (at a decentralized exchange) or supplying capital to a lending pool (at a DeFi lending platform). Agents optimally allocate their capital so that an agent invests in blockchain activities only if her payoff from doing so exceeds that of the outside option. In addition, all agents who invest in blockchain activities allocate their wealth optimally across blockchain validation and the DeFi application.

Notably, when users participate in blockchain validation or DeFi activities, their compensation is paid in terms of an asset settled on the blockchain, hereafter referred to as a cryptoasset. A consequence of receiving payments

in the cryptoasset is that, in order to consume, a user must liquidate her cryptoasset holdings through a transaction that settles directly on the blockchain. Crucially, the processing of a transaction on the blockchain incurs a delay due to congestion on the blockchain. This delay arises because the blockchain possesses a finite transaction capacity that implies that all transactions cannot be processed instantaneously. We assume that agents face heterogeneous costs of waiting and, as in practice, can pay competitive fees to have their transactions prioritized since blockchain validators endogenously accept transactions in descending fee order.

To study security, we assume that the blockchain is subject to an attack in each period from a malicious agent, hereafter referred to as an attacker. The attacker is deep pocketed and possesses an ex ante random benefit, drawn each period, from successfully disrupting the transaction activity of the blockchain. After learning her realized benefit, the attacker determines the optimal resources to expend in that period (if any) to perform such an attack. We assume that a successful attack in any period renders the blockchain inoperable thereafter. Therefore, any user with cryptoasset holdings at the time of an attack loses the ability to liquidate those holdings and hence forgoes any associated consumption utility. In turn, the payoff for investing in blockchain activities, and thus the decision to adopt the blockchain, depends on not only the cost of waiting and of paying fees but also the probability that the blockchain is compromised by an attack.

Successfully attacking the blockchain entails the attacker disrupting the blockchain so that new transactions cannot be processed, implying that any user with cryptoasset holdings cannot liquidate those holdings for consumption. Importantly, the PoW and PoS protocols specify different conditions under which a block of transactions is added to the blockchain. This implies that PoW and PoS blockchains entail different conditions for an attacker to successfully disrupt the blockchain and thus different conditions to ensure security. To provide more detail, the PoW protocol allocates the right to add a block of transactions to the blockchain to any agent who solves a computationally expensive puzzle. Agents who attempt to solve this puzzle are known as miners and their attempts to solve the puzzle are known as mining. In contrast, the PoS protocol allocates the right to add a block to the blockchain based on a lottery among a set of cryptoasset holders who agree not to sell their cryptoasset in a given period. The agents who partake in this activity are known as stakers and the act of holding cryptoasset dormant to be eligible for the lottery is known as staking.

Notably, mining and staking represent the blockchain validation activities for PoW and PoS blockchains, respectively. In either case, disrupting the blockchain requires the attacker to expend a sufficient level of resources relative to the level of investment by users into these validation activities. More specifically, for a PoW blockchain, both the attacker and the users investing in validation activities invest in specialized hardware that is used to solve the

computational mining puzzle. Moreover, the **attacker** successfully **disrupts** the **PoW** blockchain only if she **expends at least as much as the sum of all user investment into mining** (i.e., the amount of capital invested by users to solve the computational puzzle). In contrast, for a **PoS** blockchain, both the attacker and the users investing in validation activities invest in the PoS cryptoasset and hold it dormant as part of the staking process. Additionally, for a PoS blockchain, the **attacker** successfully **disrupts** the blockchain only if she **purchases** and stakes a **sufficient share** of the overall staked investment where the minimal sufficient share varies by PoS implementation but is never more than one-half and is frequently equal to one-third in modern PoS protocols.<sup>1</sup>

Ultimately, we provide two sets of results. Our first set of results, provided in [Section 3](#), compares **equilibrium security** for PoW and PoS blockchains of arbitrary scale. Our second set of results, provided in [Section 4](#), then **compares** equilibrium security for PoW and PoS blockchains specifically when the **scale** of the blockchain **grows large**. Our first set of results demonstrates that, under **current real-world conditions**, **PoS blockchains generate higher security than PoW blockchains**. Our second set of results then clarifies that the security **advantage** of PoS relative to PoW is particularly **salient at high-scale**.

To understand our first set of results, it is first important to understand two components that factor into the security of PoW versus PoS blockchains: (1) the **relative investment by the validators securing the blockchain** and (2) the **relative share of validator investment needed to successfully attack the blockchain**. We formally demonstrate that the first component always favors PoS so that PoS blockchains generate higher validation investment than PoW blockchains, all else equal. In contrast, the second component always favors PoW in that PoW requires a weakly larger share of validation investment to successfully attack the blockchain. This trade-off allows us to provide a theoretical condition that establishes when a PoW blockchain is more secure than a PoS blockchain and vice versa. However, the full range of model parameters is not realistic given current real-world conditions and thus we reexamine our findings while restricting to realistic parameters as guided by the prior literature and public data sources. More precisely, we reduce the aforementioned condition for when a PoW blockchain is more secure than a PoS blockchain to a sufficient condition that depends only on a small subset of model parameters. Then, using public data and academic sources, we identify the plausible range for those

<sup>1</sup> To provide more detail, some PoS protocols allocate the authority to add a block sequentially to a single validator, whereas other PoS protocols select a committee of validators to determine the next block. The former type of PoS protocols, known as Longest-Chain protocols, require a simple majority of staked investment to disrupt the blockchain. In contrast, the latter type of PoS protocols, generally known as Byzantine Fault Tolerant protocols, require a supermajority of more than two-thirds committee members to agree on a block for the block to be included in the blockchain (for details, see [Halaburda, He, and Li 2021](#)). Note that, in this latter case, an attacker can disrupt the PoS blockchain by acquiring only one-third of the staked investment because such a minority is sufficient to deny a supermajority coalition exceeding two-thirds of the committee. Separately, some prominent PoS protocols employ a hybrid of Longest-Chain and Byzantine Fault Tolerance (e.g., Ethereum). To capture the set of PoS protocols broadly, we treat the share of stake needed to disrupt the blockchain as a parameter, and we allow this parameter to vary to capture different PoS protocol designs.

model parameters. Notably, we demonstrate that our sufficient condition is met under any realistic parameter values. Thus, the PoS blockchain's advantage in securing higher investment is more significant than its deficit of a lower share needed to successfully attack a PoS blockchain. In turn, our first set of results, reported in [Section 3](#), finds that, all else equal, PoS blockchains are generically more secure than PoW blockchains for any scale given real-world conditions.

To provide some intuition regarding why PoS generates higher security than PoW under real-world conditions, it is important to recognize that PoW mining entails investment in mining hardware that experiences significant depreciation over time. In turn, while PoS validators require compensation for the opportunity cost of foregoing alternative investments, PoW validators require a strictly higher level of compensation for their validation services. In particular, PoW validators require compensation for both the opportunity cost of foregoing alternative investments and also for the depreciation cost of the mining hardware in which they invest. Crucially, we find that the need for elevated compensation for a PoW validator relative to a PoS validator implies a higher discount rate for validation revenues under PoW relative to PoS. This higher PoW discount rate on validation revenues implies a lower PoW validation investment level relative to the PoS investment level (all else equal). Notably, mining depreciation rates are large in practice and PoS's advantage of higher validation investment relative to PoW is thus commensurately large. We show that this advantage for PoS is sufficient to dominate PoS's disadvantages relative to PoW, yielding our result that PoS possesses higher equilibrium security than PoW under real-world conditions.

To understand our second set of results, which establishes that PoS's aforementioned advantage is particularly salient at high-scale, it is necessary to recognize that scaling has two effects on equilibrium security. The first effect is that scaling alleviates congestion and thereby endogenously reduces user fees. The lower user fees, all-else-equal, then reduce validator revenues and thus endogenously reduce equilibrium security. When this effect is dominant, then sufficient scaling can render a blockchain entirely insecure in equilibrium. The second effect is that scaling can increase user demand for the blockchain's cryptoasset due to faster processing times, thereby increasing the cryptoasset value. Thus, under this second effect, scaling increases the value of block rewards and therefore the value of validator revenues. In contrast to the first effect, this second effect can enhance blockchain security because it increases validator revenues that incentivize higher levels of validator investment. Whether the first or the second effect dominates depends on the protocol and the attacker's maximal benefit from attacking the blockchain. Our second set of results, reported in [Section 4](#), demonstrates that the first effect, a negative effect on security, dominates for a larger set of attacker benefits under a PoW protocol relative to a PoS protocol. Moreover, the second effect, a positive effect for security, dominates for a larger set of attacker benefits under a PoS

protocol relative to a PoW protocol. Our referenced results thus highlight a scaling advantage that a PoS blockchain possesses over an otherwise equivalent PoW blockchain.

Our paper relates to a large literature that studies the economics of blockchain protocols. The interested reader may consult [John, O'Hara, and Saleh \(2022\)](#) and [John and Saleh \(2025\)](#) for a survey of that literature. Particular papers within the literature that are especially related include papers examining blockchain adoption (e.g., [Cong, Li, and Wang 2021](#); [Hinzen, John, and Saleh 2022](#)), papers studying blockchain consensus generation (e.g., [Biais et al. 2019](#); [Halaburda, He, and Li 2021](#); [Saleh 2021](#)), papers studying blockchain attacks (e.g., [Chiu and Koepl 2022](#), [Pagnotta 2022](#) and [Garratt and van Oordt 2023](#)) and papers modeling key blockchain microstructure elements (e.g., [Chiu and Koepl 2022](#); [Easley, O'Hara, and Basu 2019](#); [Huberman, Leshno, and Moallemi 2021](#); [Basu et al. 2023](#)). Our work builds on the prior literature by providing the first comparative economic analysis across the two most prominent blockchain protocols, PoW and PoS.

## 1. Model

We model an infinite horizon, discrete-time setting with periods  $t \in \mathbb{N}$ . The economy is populated by overlapping generations of agents and only one asset, a cryptoasset, which is settled on a digital ledger known as a blockchain. Each agent possesses a unit endowment only in her first period and receives utility from consumption only in her last period. Each agent has access to an outside option that enables her to transfer her endowment from her first to her last period at an exogenous rate. Alternatively, the agent may invest in the blockchain, trading her endowment for the cryptoasset during her first period and then trading her cryptoasset holdings for consumption goods during her last period. Buying or selling the cryptoasset requires transacting on the blockchain, which faces particular security risks depending on whether the underlying protocol is PoW or PoS.

### 1.1 Users

Each period  $t$  begins with a unit measure of generation- $t$  agents arriving. We refer to each individual agent from generation  $t$  as Agent  $(i, t)$  with  $i \in [0, 1]$  denoting the unique identifier for that agent within the generation. Agent  $(i, t)$  lives for three periods:  $t, t+1$ , and  $t+2$ . Moreover, she is endowed with one unit of consumption goods only in her first period,  $t$ , and accrues utility from consumption only in the terminal period of her life,  $t+2$ . Each Agent  $(i, t)$  has access to an outside option with strictly positive net return  $\sigma \geq 0$ , so that applying her unit endowment to the outside option yields  $1 + \sigma$  consumption in period  $t+2$ . Alternatively, Agent  $(i, t)$  may use the blockchain, trading her endowment for units of cryptoasset during period  $t$  and then selling those units of cryptoasset and any associated proceeds for consumption goods in period  $t+2$ .

We refer to agents that utilize the blockchain as *users* and say that the agent *adopts* the blockchain technology when she utilizes the blockchain over the outside option.

We denote Agent  $(i, t)$ 's utility as  $\mathcal{U}_{(i,t)}^p$  with  $p \in \{PoW, PoS\}$  denoting the blockchain's protocol. Following the prior discussion, Agent  $(i, t)$ 's utility is given as follows:

$$\mathcal{U}_{(i,t)}^p = \max\{U_{(i,t)}^p, 1 + \sigma\}, \quad (1)$$

where  $U_{(i,t)}^p$  denotes the expected utility of Agent  $(i, t)$  if she adopts the blockchain.

To provide more detail on  $U_{(i,t)}^p$ , we assume that each user decides between investing in a DeFi protocol or engaging in *validation activity* in the intermediate period of her life. This decision is made in the intermediate period of an agent's life because the agent uses her first period to acquire cryptoassets and the last period to liquidate and consume. To elaborate further, we assume the existence of a DeFi application on the blockchain (e.g., a decentralized exchange or lending market) where the agent can invest the blockchain cryptoasset to earn an exogenous per period gross return of  $1 + \psi$ .<sup>2</sup> In contrast, validation activities relate to users competing to add blocks to the blockchain, which represents a necessary service that sustains the security and functioning of the blockchain. These validation activities — referred to as mining in a PoW blockchain and staking in a PoS blockchain — generate revenues that are shared among the users who engage in validation services. We assume that each user optimally splits her wealth between the DeFi investment and the validation investment in order to maximize the expected return from these activities. More explicitly, Agent  $(i, t)$  solves the following problem:

$$\omega_{(i,t)}^p = \arg\max_{\omega: \omega \in [0,1]} \omega \cdot \mathbb{E}_t[R_t^p] + (1 - \omega) \cdot \mathbb{E}_t[R_t^{DeFi}], \quad (2)$$

where  $R_t^{DeFi}$  denotes the period  $t$  to  $t+2$  return for users who invest in the DeFi application if the blockchain is not successfully attacked in period  $t+2$  or beforehand while  $R_t^p$  denotes the period  $t$  to  $t+2$  return for users who invest in validation activity for a blockchain governed by protocol  $p \in \{PoW, PoS\}$  if the blockchain is not successfully attacked in period  $t+2$  or beforehand. Note that  $\omega_{(i,t)}^p$  denotes the fraction of wealth that Agent  $(i, t)$  allocates towards validation activities if the blockchain is governed by protocol  $p \in \{PoW, PoS\}$  and if Agent  $(i, t)$  adopts the blockchain. In turn, if the blockchain is not successfully attacked in period  $t+2$  or beforehand, the period  $t$  to  $t+2$  return from adopting the cryptoasset and optimally investing across DeFi and validation activities is given by:

$$\Pi_{(i,t)}^p = \omega_{(i,t)}^p \cdot R_t^p + (1 - \omega_{(i,t)}^p) \cdot R_t^{DeFi} \quad (3)$$

<sup>2</sup> We impose  $\psi > \sigma$  because, otherwise, the outside option is always preferred to blockchain investment and thus the model solution is trivially a zero adoption and zero security equilibrium for both PoW and PoS.

Then, since Agent  $(i, t)$  possesses a unit endowment, she receives  $\Pi_{(i,t)}^p$  consumption goods in period  $t+2$  for liquidating her cryptoasset holdings when she adopts the blockchain in period  $t$  and is able to liquidate her holdings in period  $t+2$ . Nonetheless, Agent  $(i, t)$  does not necessarily receive  $\Pi_{(i,t)}^p$  in period  $t+2$  when she adopts the blockchain in period  $t$  and thus Agent  $(i, t)$ 's decision to adopt blockchain technology involves additional concerns. First, the blockchain may not be able to process transactions because the blockchain may be successfully attacked on or before period  $t+2$ . Crucially, a successful attack renders the blockchain inoperable in that period and thereafter so that User  $(i, t)$  forgoes her return  $\Pi_{(i,t)}^p$  if there is a successful attack on or before period  $t+2$ . We discuss this concern in detail in Section 1.3. Second, even if the blockchain is not successfully attacked, Agent  $(i, t)$  may not receive immediate processing because the blockchain possesses a finite transaction rate and therefore is subject to congestion.

To incorporate the effects of congestion, we assume that each Agent  $(i, t)$  possesses utility over period  $t+2$  consumption and an intraperiod wait disutility during that period. Further, as alluded to earlier, Agent  $(i, t)$  may pay a fee,  $f_{(i,t)}^p \geq 0$ , denominated in consumption goods, to reduce her wait time when liquidating in the last period of her life. Agent  $(i, t)$  is willing to pay this fee because the blockchain optimally processes transactions in descending fee order. Denote by  $c_{(i,t)}$  Agent  $(i, t)$ 's wait disutility per unit time and denote by  $W^p(f, f_{-(i,t)})$  the expected wait time of Agent  $(i, t)$  when she pays fee  $f$  and the other users pay fees  $f_{-(i,t)}$ . Then, Agent  $(i, t)$ 's total disutility from waiting equals  $c_{(i,t)} \cdot W^p(f, f_{-(i,t)})$ . We assume that  $c_{(i,t)}$  is drawn from a smooth cumulative distribution,  $G \in \mathcal{C}^\infty[0, \infty)$ , with a non-negative support and a finite first moment (i.e.,  $\int_0^\infty c \, dG(c) < \infty$ ). If Agent  $(i, t)$  does not use the blockchain then she optimally pays a fee  $f_{(i,t)}^p = 0$ ; otherwise, she selects her fee to maximize period  $t+2$  consumption,  $\Pi_{(i,t)}^p$ , net the fee and total wait disutility, which amounts to choosing  $f_{(i,t)}^p$  to solve:

$$f_{(i,t)}^p = \arg \max_{f: f \geq 0} \underbrace{\Pi_{(i,t)}^p - f}_{\text{Consumption}} - \underbrace{c_{(i,t)} \cdot W^p(f, f_{-(i,t)})}_{\text{Wait disutility}} \quad (4)$$

Note that  $f_{(i,t)}^p$  is a function of Agent  $(i, t)$ 's wait disutility  $c_{(i,t)}$  and the beliefs Agent  $(i, t)$  has regarding the fees  $f_{-(i,t)}^p$  chosen by the other agents. We do not explicitly reference this dependence in what follows in order to ease the notation.

Let  $\pi_{t \rightarrow t+2}^p \in [0, 1]$  denote the probability that the blockchain survives (i.e., is not successfully attacked) until the end of period  $t+2$  conditional on surviving until the beginning of period  $t$ . Then, the expected utility of Agent  $(i, t)$  from using the blockchain — that is, purchasing cryptoasset units in period  $t$  and selling her cryptoasset holdings in  $t+2$  while paying fee  $f_{(i,t)}^p$  — is given by:

$$U_{(i,t)}^p = \pi_{t \rightarrow t+2}^p \cdot \mathbb{E}_t[\Pi_{(i,t)}^p - f_{(i,t)}^p - c_{(i,t)} W^p(f_{(i,t)}^p, f_{-(i,t)}) \mid c_{(i,t)}] \quad (5)$$

where we assume that the User  $(i, t)$  receives zero consumption utility in the case that a successful attack occurs before she liquidates her period  $t+2$  cryptoasset holdings. We use  $\mathbb{E}_t[\cdot]$  to denote an expectation conditional on all public information available at the beginning of period  $t$ . Note that the budget constraint of Agent  $(i, t)$  is represented by  $P_t^P \cdot Q_{(i,t),t}^P \leq 1$ , which states that the cost of the cryptoasset that Agent  $(i, t)$  purchases cannot exceed her initial endowment. We hereafter restrict attention to the case in which each agent either invests all of her wealth on the blockchain or invests all her wealth through the outside option. This assumption is without loss of generality because splitting wealth between blockchain investment and the outside option is never optimal (except for the marginal user who is indifferent).

## 1.2 Blockchain

A blockchain is an electronic ledger that records transactions in discrete chunks referred to as *blocks*. The blocks are concatenated together into a single *chain*, hence the term blockchain. For the blockchain to function, there must be some agents responsible for creating the blocks because transactions enter the blockchain only by being recorded on blocks that are added to the chain. We let  $\Lambda > 0$  denote the blockchain's transaction rate, which is the rate at which the blockchain accepts transactions. To avoid unnecessary complications, we assume that block sizes are *small* in the sense that transactions are continuously accepted to the blockchain in infinitesimally small blocks.<sup>3</sup> This enables us to derive the following expression for the expected wait time,  $W^P(f, f_{-(i,t)})$ :

$$W^P(f, f_{-(i,t)}) = \underbrace{\frac{1}{\Lambda}}_{\text{Service time per user}} \times \underbrace{\int \mathbb{1}\{f_{(j,t)}^P \geq f\} dG(c_{(j,t)})}_{\text{Higher-paying users}} \quad (6)$$

The agents that provide the service of creating blocks are generally known as *validators* but, as discussed earlier, are more specifically referred to as miners for PoW blockchains and stakers for PoS blockchains. In either case, validators receive compensation for creating blocks. That compensation arises in two forms: fees and block rewards. As discussed in Section 1.1, fees refer to the user payments  $f_{(i,t)}^P$  denominated in the consumption good. Block rewards refer to newly created units of the cryptoasset, which we occasionally refer to as coins. The coins are distributed into circulation by giving them as rewards to the validators who create new blocks, hence the term block reward. We assume that these block rewards are distributed according to a constant cryptoasset supply growth rate,  $\rho \geq 0$ . Explicitly, denoting by  $M_t$  the units of cryptoasset outstanding at the beginning of period  $t$ , we have that:

<sup>3</sup> In principle, specifying the blockchain's transaction rate,  $\Lambda$ , allows for an arbitrary block size,  $b$ , because the specified transaction rate is achieved by a block arrival rate of  $\frac{\Lambda}{b}$ . Formally, our analysis considers the limit case when  $b \rightarrow 0^+$  because arbitrary block sizes complicate the derivation of the wait time without providing incremental economic insight.

$$M_{t+1} = M_t e^\rho \quad (7)$$

As a normalization, we assume that the initial cryptoasset supply is given by  $M_0 = 1$ . In that case, the total block reward distributed across period  $t$ , denoted by  $B_t$ , is then given by:

$$B_t \equiv M_{t+1} - M_t = M_t(e^\rho - 1) = e^{\rho t}(e^\rho - 1). \quad (8)$$

We assume that the block reward  $B_t$  is distributed uniformly across blocks in period  $t$ . Additional details regarding the blockchain vary by protocol, so we detail the PoW and PoS protocols separately hereafter.

**1.2.1 PoW blockchain.** A PoW blockchain accepts a new block proposed by a miner only if that block contains the solution to a prespecified computational puzzle. To find the solution for such a puzzle, a miner born in period  $t$  must purchase computational hardware in period  $t$  and then expend the associated computational power in the intermediate period of her life  $t+1$ . We measure computational power in units known as *hash power*, and we refer to the total hash power expended by all users in period  $t+1$  as  $H_{t+1}$ . Miners are willing to bear this computation expense only because they receive compensation for providing this service. As already discussed, miners receive compensation via block rewards and fees, which are shared *pro rata* among miners with respect to the amount of hash power expended by each miner. Recall that, the total amount of block rewards paid in period  $t+1$  is given by  $B_{t+1}$ . Moreover, the value of fees paid in period  $t+1$  is the sum of fees paid by agents born in period  $t-1$  who are liquidating their holdings in the terminal period of their life,  $t+1$ , and therefore given by  $\int f_{(i,t-1)}^p dG(c_{(i,t-1)})$ .

We assume that the timing of the model is such that each user must pay mining costs upfront in the first period of her life in order to mine in the second period of her life. Finally, after mining in the second period of her life, each user liquidates her cryptoasset holdings (i.e., accrued block rewards in addition to any cryptoassets that may have been purchased). As in practice, we assume that mining costs arise from two sources: (1) a cost  $\kappa_h$  per unit of hash power,  $H_{t+1}$ , on energy expenditure and (2) a cost for mining equipment, which we normalize to 1 per unit of hash power generated by the mining equipment. We allow for the possibility that mining equipment has some residual value after use and thus each miner also liquidates her mining equipment in the last period of her life to gain this residual value. More formally, conditional on survival of the blockchain, miners are able to liquidate their mining equipment for its initial value minus a proportional factor,  $\delta \in [0, 1]$ , which represents depreciation of the mining equipment (e.g., occurring due to advances in mining technology and/or general wear and tear). Thus, generating hash power  $H_{t+1}$  requires an upfront cost of  $H_{t+1} \cdot (1 + \kappa_h)$  consumption goods and conditional on blockchain survival the miner redeems  $H_{t+1} \cdot (1 - \delta)$  in consumption goods from liquidating her mining equipment

in the last period of her life. As a consequence, if the blockchain is not successfully attack in period  $t+2$  or beforehand, then the lifetime return for a blockchain user born in period  $t$  who mines in period  $t+1$  is given as follows:

$$R_t^{PoW} = \frac{H_{t+1} \cdot (1 - \delta) + (B_{t+1} + \frac{\int f_{(i,t-1)}^{PoW} dG(c_{(i,t-1)})}{P_{t+1}^{PoW}}) \cdot P_{t+2}^{PoW}}{H_{t+1} \cdot (1 + \kappa_h)} \quad (9)$$

To understand Equation (9), note that  $H_{t+1}(1 + \kappa_h)$  is the total investment into mining activities from users born in period  $t$ . Further,  $(B_{t+1} + \frac{\int f_{(i,t-1)}^{PoW} dG(c_{(i,t-1)})}{P_{t+1}^{PoW}}) \cdot P_{t+2}^{PoW}$  denotes the period  $t+2$  value of all block rewards and fees earned from mining in period  $t+1$ . Importantly, fees are denominated in consumption value but must be paid in units of the underlying cryptoasset. Therefore, the cumulative fees paid in period  $t+1$  are paid in the form of  $\frac{\int f_{(i,t-1)}^{PoW} dG(c_{(i,t-1)})}{P_{t+1}^{PoW}}$  units of cryptoasset in period  $t+1$ , which is then liquidated for a consumption value of  $\frac{\int f_{(i,t-1)}^{PoW} dG(c_{(i,t-1)})}{P_{t+1}^{PoW}} \cdot P_{t+2}^{PoW}$  in period  $t+2$ . In contrast, block rewards are paid in the form of  $B_{t+1}$  units of cryptoasset, which are held and liquidated for a terminal consumption value of  $B_{t+1} \cdot P_{t+2}^{PoW}$ . Finally, as mentioned above  $H_{t+1} \cdot (1 - \delta)$  represents the proceeds from selling all mining equipment at the end of period  $t+1$ . Thus,  $R_t^{PoW}$  represents the total revenues earned from mining and salvaging equipment divided by the total investment into mining by users born in period  $t$ .

**1.2.2 PoS blockchain.** A PoS blockchain involves no computational puzzle. Rather, a PoS protocol randomly selects a coin (or coins) from a set of *staked* coins, each of which has been contributed to the set by its associated coin owner. If a user places a coin into the described set, then the coin is said to have been staked, and the user is referred to as a staker. The owner(s) of selected coin(s) then participate in the creation and approval of the next block on the blockchain and, as discussed, receives compensation in the form of block rewards and fees. Crucially, a user can stake a cryptoasset only if she already owns that cryptoasset. As a consequence, a user cannot stake a cryptoasset in the first period of her life because she is not endowed with any cryptoassets, and she optimally would not stake a cryptoasset in the last period of her life because she wishes to sell her holdings and consume in that period. Stated differently, a user stakes only in the intermediate period of her life because she can acquire cryptoassets only in the first period of her life, and she does not benefit from selling her cryptoasset holdings in the intermediate period of her life because she does not value consumption in that period.

Recall that, for simplicity, we consider a limiting case whereby infinitely many blocks of infinitesimal size are added to the blockchain in each period. This implies that while the rewards and fees earned by a given staker from a

particular block is random, the total rewards and fees accrued in a period is nonetheless nonrandom. In turn, the realized rewards and fees accrued by any PoS blockchain user equals the expected rewards and fees accrued by that user (i.e., there is no aggregate risk). Thus, given these features, the rewards and fees are effectively shared *pro rata* among all users who stake their cryptoasset. For this reason, denoting the total amount of cryptoasset staked in period  $t+1$  as  $S_{t+1}$ , then the lifetime return to staking for a user born in period  $t$  who stakes in period  $t+1$  is given by

$$R_t^{PoS} = \frac{S_{t+1} \cdot P_{t+2}^{PoS} + (B_{t+1} + \frac{\int f_{(i,t-1)}^{PoS} dG(c_{(i,t-1)})}{P_{t+1}^{PoS}}) \cdot P_{t+2}^{PoS}}{S_{t+1} \cdot P_t^{PoS}} \quad (10)$$

To understand Equation (10) first note that  $S_{t+1} \cdot P_t^{PoS}$  represents the total investment from users born in period  $t$  that will be allocated to staking in period  $t+1$ . Then, as explained in Section 1.2.1,  $(\frac{\int f_{(i,t-1)}^{PoS} dG(c_{(i,t-1)})}{P_{t+1}^{PoS}}) \cdot P_{t+2}^{PoS}$  represents the period  $t+2$  consumption value derived from liquidating all fees and block rewards earned in period  $t+1$ . Further, we note that after staking, the  $S_{t+1}$  coins used for validation will be returned to the users in the terminal period of their life and liquidated for a value of  $S_{t+1} \cdot P_{t+2}^{PoS}$ . Thus,  $R_t^{PoS}$  represents the total revenues earned from staking plus the principal of the initially staked coins divided by the total investment into staking by users born in period  $t$ .

**1.2.3 DeFi investment.** As noted earlier, investing one unit of the cryptoasset into the DeFi application yields a gross return equivalent to  $1 + \psi$  units of the cryptoasset. For simplicity, we assume that DeFi revenues are paid by users that we do not model and are paid through a separate cryptoasset specific to the DeFi application. These assumptions ensure that the user payments generating the DeFi revenues do not affect the market clearing conditions for the blockchain's cryptoasset and thereby simplify our equilibrium analysis.

To provide more detail on the DeFi return, recall that a user born in period  $t$  who wishes to invest 1 unit of consumption goods in the DeFi application must first exchange her consumption goods for cryptoassets in period  $t$ , receiving  $\frac{1}{P_t}$  coins at that time. Thereafter, in period  $t+1$ , the user invests her  $\frac{1}{P_t}$  coins in the DeFi application. Moreover, since the DeFi protocol yields a gross per-period return of  $1 + \psi$ , the value of her DeFi investment increases to  $\frac{1}{P_t} \cdot (1 + \psi)$  coins at the end of period  $t+1$ . Then, since  $P_{t+2}$  corresponds to the value of each coin in period  $t+2$ , the user liquidates her holdings in period  $t+2$  for a consumption value of  $P_{t+2} \cdot \frac{1}{P_t} \cdot (1 + \psi)$ . Thus, the lifetime return of a user born in period  $t$  from investing in the DeFi application in period  $t+1$  is given by:

$$R_t^{DeFi} = \frac{P_{t+2}}{P_t} \cdot (1 + \psi). \quad (11)$$

### 1.3 Attacker

We model a malicious agent, hereafter referred to as an attacker, that seeks to sabotage the blockchain. This sabotage entails the attacker suspending transactions from being posted to the blockchain (i.e., a Denial-of-Service attack). The attacker is able to suspend transactions from being added to the blockchain by precluding other validators from adding blocks with transactions to the blockchain. As we discuss in [Section 1.3.1](#), an attacker gains the ability to disrupt the transaction activity of a PoW blockchain by acquiring a majority of the total computational power spent on mining (see, e.g., [Nakamoto 2008](#)). In contrast, the ability to disrupt transaction activity in modern PoS blockchain implementations can arise even with a minority stake, as we discuss in detail in [Section 1.3.2](#). We follow [Pagnotta \(2022\)](#) and assume that if the blockchain is successfully attacked in any period then it incurs a crisis of confidence and is no longer operable thereafter, rendering the cryptoasset and any specialized mining equipment (in the PoW case) valueless. As we will discuss subsequently, the attacker internalizes the difficulty of executing the attack and attacks only if she finds mounting an attack to be incentive compatible.

We assume that the attacker has unlimited resources to perform an attack and receives an ex ante random benefit from successfully disrupting the blockchain. Note that a PoW attack requires the purchase of significant computational equipment and power beforehand, whereas a PoS attack requires the purchase of significant PoS cryptoasset beforehand. In either case, if the attacker wants to attack the blockchain in period  $t+1$ , then she must spend resources in the prior period  $t$  to prepare for the attack. We consequently assume that the attacker learns her benefit from successfully attacking the blockchain  $\Gamma_t \sim U[0, \bar{\Gamma}]$  at the beginning of period  $t$  and thereafter chooses an amount of resources  $A_t \geq 0$ , to mount an attack also at the beginning of period  $t+1$ . We specify both  $\Gamma_t$  and  $A_t$  in terms of consumption goods and assume that  $\bar{\Gamma} > 0$  represents the maximal benefit the attacker can receive from disrupting the blockchain. Then, letting  $v_{t+1}^p(A_t)$  denote the probability that the attacker successfully attacks the blockchain at the beginning of period  $t+1$  when using resources  $A_t$  given protocol  $p \in \{PoW, PoS\}$ , the attacker's problem is given formally as follows:

$$\max_{A_t \geq 0} \Gamma_t \cdot v_{t+1}^p(A_t) - A_t \quad (12)$$

where we assume that the attacker selects  $A_t$  optimally so as to maximize her consumption utility given  $\Gamma_t$  and the equilibrium probability of a successful attack  $v_{t+1}^p(A_t)$ . We explicitly provide the probability  $v_{t+1}^p(A_t)$  for each protocol  $p \in \{PoW, PoS\}$  in the following [Sections 1.3.1](#) and [1.3.2](#), respectively.

**1.3.1 PoW attacks.** For a PoW blockchain, suspending transaction activity on the blockchain entails the attacker forking the blockchain and adding only empty blocks to her forked branch. Crucially, the attacker adds these empty

blocks in an attempt to deny all users from having their transactions processed by the blockchain. Akin to Pagnotta (2022), we assume that an attack succeeds if and only if the attacker is able to make her forked branch arbitrarily longer than the main chain.<sup>4</sup> Note that an attacker's ability to create a forked branch that is arbitrarily longer than the main chain depends on her computational power relative to all other miners. To initiate the attack, the attacker acquires the necessary computational equipment and power in period  $t$  by expending  $A_t$  and then uses that power to launch an attack at the beginning of period  $t + 1$ . We assume that the attacker faces the same cost as the users for energy expenditure and mining equipment per unit of hash power. Namely, an expense of  $A_t$  by the attacker implies an attacker hash power of  $\frac{A_t}{1+\kappa_h}$  since the attacker must pay for both the mining equipment and the energy expenditure.

If the attacker possesses (weakly) higher computational power than the other miners in period  $t + 1$  (i.e.,  $A_t \geq H_{t+1} \cdot (1 + \kappa_h)$ ), then her forked branch grows at a faster rate than the main chain. Moreover, in that case, standard results from the stochastic processes literature (see, e.g., Karlin and Taylor 1975) imply that the attacker's forked branch eventually becomes longer than the main chain with certainty so that the attack succeeds with probability one. In contrast, if the attacker possesses less computational power than the other miners in period  $t + 1$  (i.e.,  $A_t < H_{t+1} \cdot (1 + \kappa_h)$ ), then the main chain grows at a faster rate than her forked branch and therefore the likelihood that the attacker's forked branch ever exceeds the main chain by an arbitrary  $k$  blocks vanishes to zero as  $k$  diverges so that the attack fails with probability one. Thus,  $v_{t+1}^{PoW}(A_t)$  is given explicitly by:

$$v_{t+1}^{PoW}(A_t) = \begin{cases} 1 & \text{if } A_t \geq H_{t+1} \cdot (1 + \kappa_h) \\ 0 & \text{if } A_t < H_{t+1} \cdot (1 + \kappa_h) \end{cases} \quad (13)$$

Moreover, we can derive the attackers optimal expenditure and, as a consequence, the probability of a successful attack as follows:

$$A_t^*(\Gamma_t) = \begin{cases} H_{t+1} \cdot (1 + \kappa_h) & \text{if } \Gamma_t \geq H_{t+1} \cdot (1 + \kappa_h) \\ 0 & \text{if } \Gamma_t < H_{t+1} \cdot (1 + \kappa_h) \end{cases}$$

$$v_{t+1}^{PoW}(A_t^*(\Gamma_t)) = \begin{cases} 1 & \text{if } \Gamma_t \geq H_{t+1} \cdot (1 + \kappa_h) \\ 0 & \text{if } \Gamma_t < H_{t+1} \cdot (1 + \kappa_h) \end{cases} \quad (14)$$

Hence, we see that whenever the attacker's benefit from attacking the blockchain,  $\Gamma_t$ , is greater than the cost of accumulating the equilibrium hash

<sup>4</sup> Formally, we consider the probability that the attacker's chain ever exceeds the main chain by  $k \in \mathbb{N}$  blocks within a period, noting that for a sufficiently large  $k$  we expect a crisis of confidence as users and validators learn the attack is occurring. For exposition, we study the limiting case in which a crisis of confidence occurs after the attacker's fork exceeds the main chain by an arbitrary number of blocks, which is the limiting case as  $k \rightarrow \infty$ . Our main results would hold for finite  $k$  as well, but this approach simplifies the solution while preserving the same insights.

rate, that is,  $H_{t+1} \cdot (1 + \kappa_h)$ , then the attacker will optimally use sufficient resources  $A_t$  to ensure their period  $t + 1$  attack succeeds with probability one and otherwise will not find it optimal to expend any resources to attack the blockchain.

**1.3.2 PoS attacks.** For a PoS blockchain, suspending transaction activity on the blockchain entails the attacker acquiring and staking sufficiently many coins relative to the number of coins staked by other coin holders. Importantly, though, an attacker can preclude transactions from being added to a PoS blockchain even with a minority of staked coins. More specifically, within modern PoS blockchains, a sequence of blocks being added to the blockchain requires supermajority approval from a set of committees where committee composition is determined per the distribution of staked coins.<sup>5</sup> In turn, since a supermajority is necessary to add blocks to the blockchain, a complementary minority is necessarily sufficient to preclude transactions from being added to the blockchain because this minority can abstain from voting and thereby preclude the needed supermajority vote.

To provide more detail, modern PoS blockchains require the support of a supermajority  $\eta \geq \frac{1}{2}$  of stake to ensure inclusion of blocks to the blockchain. Then, as a consequence, an attack will succeed so long as the attacker stakes at least  $1 - \eta$  of the overall stake. The attacker expending initial capital  $A_t$  in period  $t$  implies that the attacker is able to purchase  $\frac{\lambda \cdot A_t}{P_t^{PoS}}$  PoS coins, where  $\lambda \geq 1$  incorporates the potential leverage of the attacker. In particular, we incorporate cryptoasset leverage for the attacker when studying PoS blockchains (i.e.,  $\lambda \geq 1$ ) because asset lending markets, if sufficiently developed in the future, could facilitate more PoS cryptoasset lending when compared to PoW. Of particular note, in our practical analysis in we discuss how  $\lambda - 1$  can be interpreted as PoS native cryptoasset leverage relative to PoW native cryptoasset leverage (see [Remark 2](#) in [Section 3.2](#)). Recall that we denote the number of staked coins in period  $t + 1$  by  $S_{t+1}$  and thus an attack mounted with capital  $A_t$  succeeds if and only if:

$$\underbrace{\frac{\lambda \cdot A_t}{P_t^{PoS}}}_{\text{Attacker Stake in Period } t+1} \geq \underbrace{(1 - \eta)}_{\text{Share Needed To Disrupt}} \times \underbrace{\left(S_{t+1} + \frac{\lambda \cdot A_t}{P_t^{PoS}}\right)}_{\text{Overall Stake in Period } t+1} \quad (15)$$

<sup>5</sup> The specific details of modern PoS protocols are beyond the scope of this paper, but the interested reader may consult [Schwarz-Schilling et al. \(2023\)](#) for details regarding the most widely used PoS protocol, Ethereum. Regardless, to provide further context, some leading PoS protocols determine the number of votes that a user receives on a given committee according to a binomial distribution with  $n$  trials and a trial success probability  $p_{(i,t)}$ , which varies by user  $(i,t)$ . In this context,  $n$  is specified as the (expected) number of committee members and  $p_{(i,t)}$  is specified as the proportion of staked capital from user  $(i,t)$  among all users. Crucially, a sequence of blocks being added to the blockchain requires a supermajority *across* committees and there is at least one committee per block. Then, since we assume infinitely many blocks per epoch, the distribution of membership across all committees in an epoch aggregates to the distribution of stake among users. As a consequence, a supermajority of committee votes across the epoch is equivalent to a supermajority of staked capital within the epoch.

Further, denoting by  $\mathcal{S}_{t+1} = S_{t+1} P_t^{PoS}$  the amount of capital utilized in period  $t$  to purchase coins that will be staked in period  $t+1$ , note that Equation (15) can be rewritten as follows:

$$A_t \geq \frac{1-\eta}{\lambda \cdot \eta} \times S_{t+1} P_t^{PoS} \equiv \alpha \cdot \mathcal{S}_{t+1}, \quad (16)$$

where we define  $\alpha := \frac{1-\eta}{\lambda \cdot \eta} \in [0, 1]$  for exposition. Hereafter, we refer to  $\alpha$  as the *relative cost* of attacking the PoS blockchain since the implication of Equation (16) is that for every unit of staked capital, the attacker needs to purchase and stake  $\alpha \leq 1$  units of capital in order to successfully attack the blockchain. Given the prior discussion, the probability of a successful attack  $v_{t+1}^{PoS}(A_t)$  given an amount of initial capital  $A_t$  spent by the attacker to execute an attack in period  $t+1$  is given explicitly by:

$$v_{t+1}^{PoS}(A_t) = \begin{cases} 1 & \text{if } A_t \geq \alpha \cdot \mathcal{S}_{t+1} \\ 0 & \text{if } A_t < \alpha \cdot \mathcal{S}_{t+1} \end{cases} \quad (17)$$

Equation (17) formalizes that the security features of the PoS blockchain imply that whenever expending  $A_t \geq \alpha \cdot \mathcal{S}_{t+1}$ , then the attacker stakes enough coins in period  $t+1$  relative to the users to ensure that the attack succeeds in period  $t+1$  with certainty, implying that the blockchain survives with probability zero. In contrast, when expending  $A_t < \alpha \cdot \mathcal{S}_{t+1}$ , then the period  $t+1$  attack fails with certainty and the blockchain survives with probability one. This comes from the fact that  $\mathcal{S}_{t+1}$  represents the total number of coins purchased by users born in period  $t$  that are staked in period  $t+1$ . Then, per Equations (15) and (16), the attacker would need to spend  $A_t \geq \alpha \cdot \mathcal{S}_{t+1} P_t^{PoS} = \alpha \cdot \mathcal{S}_{t+1}$  in period  $t$  to acquire a sufficient fraction of stake,  $1-\eta$ , to ensure that the period  $t+1$  attack succeeds. Given these features, the optimal expenditure and success probability of the attacker in a PoS blockchain are given by:

$$\begin{aligned} A_t^*(\Gamma_t) &= \begin{cases} \alpha \cdot \mathcal{S}_{t+1} & \text{if } \Gamma_t \geq \alpha \cdot \mathcal{S}_{t+1} \\ 0 & \text{if } \Gamma_t < \alpha \cdot \mathcal{S}_{t+1} \end{cases} \\ v_{t+1}^{PoS}(A_t^*(\Gamma_t)) &= \begin{cases} 1 & \text{if } \Gamma_t \geq \alpha \cdot \mathcal{S}_{t+1} \\ 0 & \text{if } \Gamma_t < \alpha \cdot \mathcal{S}_{t+1} \end{cases} \end{aligned} \quad (18)$$

Namely, if  $\Gamma_t \geq \alpha \cdot \mathcal{S}_{t+1}$  then the attacker optimally launches an attack with sufficient resources,  $A_t$ , to ensure that the attack succeeds with probability one, and otherwise does not mount an attack because doing so is not incentive compatible.

## 1.4 Equilibrium

Before proceeding to define the equilibrium of the model, we will first summarize the general timing of the model both from the perspective of

(a) the activity of a single user over their lifetime and (b) the perspective of all activity that occurs within a single period. Namely, from the perspective of a particular user, the timing of the model is as follows. User  $(i, t)$  is born in period  $t$  after which she decides whether or not to adopt the blockchain. Conditional on adopting the blockchain, the user then allocates her endowment wealth between DeFi and validation activities. To invest in DeFi activities, User  $(i, t)$  purchases cryptoasset in period  $t$  from users that are liquidating in the terminal period of their life. To invest in validation activities, User  $(i, t)$  purchases either mining equipment in period  $t$  (for a PoW blockchain) or purchases additional cryptoasset to stake (for PoS blockchain). Next, in period  $t+1$ , User  $(i, t)$  provides the DeFi investment and validation services using the cryptoasset and applicable equipment purchased in period  $t$  and earns any rewards from doing so. Finally, in period  $t+2$ , User  $(i, t)$  liquidates any cryptoasset holdings, purchased or earned in prior periods, for consumption by selling to newly arriving users that are born in period  $t+2$ . At the end of period  $t+2$ , User  $(i, t)$  exits the model.

Next we will summarize the timing of all activity that occurs within a particular period  $t$ . Namely, at the start of each period  $t$  the attacker attacks the blockchain if they had mounted sufficient resources to do so in the previous period  $t-1$ . Conditional on no successful attack, the measure of users born in period  $t$  that adopt the blockchain purchase the necessary cryptoasset for investment into the DeFi protocol as well as any equipment (PoW) or cryptoasset (PoS) needed to perform validation activities. All cryptoasset purchased by users born in period  $t$  is sold to them by users born in period  $t-2$  who are liquidating their cryptoasset holdings (both purchased and earned) for consumption in the terminal period of their life. When users born in period  $t-2$  liquidate their cryptoasset holdings, they do so by recording a transaction on the blockchain, facilitated by the period  $t$  validators who are users that were born in period  $t-1$ . The users liquidating their cryptoasset holdings in period  $t$  pay fees in period  $t$  to the validators who record their transactions on the blockchain. Finally, at the start of period  $t$ , the attacker learns her benefit from successfully attacking the blockchain in the subsequent period and decides whether or not to mount an attack for that period.

Akin to [Huberman, Leshno, and Moallemi \(2021\)](#) and [Hinzen, John, and Saleh \(2022\)](#), we restrict ourselves to examining a stationary cutoff equilibrium, characterized by an endogenously determined adoption cutoff  $c^p$  such that Agent  $(i, t)$  adopts the blockchain technology (over the outside option) if and only if  $c_{(i,t)} < c^p$ . Furthermore, we suppose that all agents utilize a symmetric ex ante fee strategy  $\phi^p$ , which maps each user's realized cost of waiting  $c$  to the fee they pay  $f = \phi^p(c)$ . For regularity, we impose that  $\phi^p$  is twice continuously differentiable on  $(0, c^p)$  and both continuous and strictly increasing on  $[0, c^p]$ .

Within our model, a  $p \in \{PoW, PoS\}$  equilibrium is (1) an adoption cutoff,  $c^p$ , (2) a function,  $\phi^p(c)$ , that maps user types to their fees, (3) a set of fee

realizations  $\{f_{(i,t)}^p\}_{(i,t):i \in [0,1], t \geq 0}$  such that  $f_{(i,t)}^p \equiv \phi^p(c_{(i,t)})$  for each Agent  $(i, t)$ , (4) a cryptoasset market value,  $\mathcal{M}^p$ , (5) the fraction of wealth that each user invests in validation activities  $\omega^p$ , (6) a probability  $\pi^p$  indicating the likelihood that the blockchain will not be attacked within an agent's lifetime assuming that the blockchain has not already been successfully attacked, and (7) for PoW (a) the total mining computational power,  $H^*$ , and for PoS (b) the amount of capital utilized in each period  $t$  to purchase coins that will be staked in period  $t+1$ ,  $\mathcal{S}^*$ . All described quantities are conditional on blockchain survival until the relevant period. After a successful blockchain attack, the blockchain is not viable, so no user adopts the blockchain. The equilibrium is defined by the following conditions:

- (i) Blockchain adoption decisions are optimal Agent  $(i, t)$  adopting the blockchain entails her selling her endowment for cryptoasset. More precisely, given the nature of the cut off equilibrium with threshold  $c^p$ , all agents adopt whenever  $c_{(i,t)} < c^p$ . Thus, the optimality of adoption in any cutoff equilibrium requires that for all  $(i, t)$ :

$$c_{(i,t)} < c^p \Leftrightarrow U_{(i,t)}^p > 1 + \sigma \quad (19)$$

Whenever Agent  $(i, t)$  adopts the blockchain they invest their full wealth and therefore it must be the case that they purchase  $\frac{1}{P_t^p}$  coins with  $P_t^p \equiv \frac{\mathcal{M}^p}{M_t}$  defined as the price of the cryptoasset in period  $t$ , and  $M_t = e^{\rho t}$  being the units of cryptoasset outstanding at the beginning of period  $t$ . In addition, note that our cutoff equilibrium implies that the total wealth spent on purchasing cryptoasset in each period  $t$  is given by  $G(c^p)$  because:

$$\text{For all } t: |\{(i, t): U_{(i,t)}^p > 1 + \sigma\}| = |\{(i, t): c_{(i,t)} < c^p\}| = G(c^p) \quad (20)$$

- (ii) Equilibrium fees are optimal We require that Agent  $(i, t)$  with realized cost  $c_{(i,t)} \in [0, \infty)$  finds it optimal to pay the fee  $f_{(i,t)}^p = \phi^p(c_{(i,t)})$  given that all other agents  $(j, t) \neq (i, t)$  pay fees according to  $f_{(j,t)}^p = \phi^p(c_{(j,t)})$ . Formally, the following condition holds:

For all

$$c: \phi^p(c) = \begin{cases} \argmax_{f: f \geq 0} \Pi_{i,t}^p - f - \frac{c}{\Lambda} \int \mathbb{1}\{f_{(j,t)}^p \geq f\} dG(c_{(j,t)}) & \text{if } c < c^p \\ 0 & \text{if } c \geq c^p \end{cases} \quad (21)$$

which uses the fact that agents optimally pay zero fees whenever they do not adopt.

- (iii) Users optimally allocate wealth to validation activities In a  $p \in \{PoW, PoS\}$  equilibrium, we assume that users symmetrically allocate

their wealth between validation (mining in PoW or staking in PoS) and DeFi investment. More formally, we impose  $\omega_{(i,t)}^p = \omega^p$  for all  $(i, t)$ , where

$$\omega^p = \arg \max_{\omega \in [0,1]} \omega \cdot \mathbb{E}_t[R_t^p] + (1 - \omega) \cdot \mathbb{E}_t[R_t^{DeFi}],$$

where the returns  $R_t^{PoW}$ ,  $R_t^{PoS}$ , and  $R_t^{DeFi}$  are given as:

$$R_t^{PoW} = \frac{H_{t+1} \cdot (1 - \delta) + (B_{t+1} + \frac{\int f_{(i,t-1)}^{PoW} dG(c_{(i,t-1)})}{P_{t+1}^{PoW}}) \cdot P_{t+2}^{PoW}}{H_{t+1} \cdot (1 + \kappa_h)} \quad (22)$$

$$R_t^{PoS} = \frac{S_{t+1} \cdot P_{t+2}^{PoS} + (B_{t+1} + \frac{\int f_{(i,t-1)}^{PoS} dG(c_{(i,t-1)})}{P_{t+1}^{PoS}}) \cdot P_{t+2}^{PoS}}{S_{t+1} \cdot P_t^{PoS}} \quad (23)$$

$$R_t^{DeFi} = \frac{P_{t+2}^p}{P_t^p} \cdot (1 + \psi), \quad \psi > \sigma, \quad p \in \{PoW, PoS\} \quad (24)$$

- (iv) The cryptoasset market clears For each period  $t$ , the total user demand and supply for cryptoasset units varies by protocol.

PoW: The demand for a PoW cryptoasset in period  $t$  is given by the fraction of investors that invest in the DeFi option times the number of coins that each investor can purchase with their endowment given as  $(1 - \omega^{PoW}) \cdot \frac{G(c^{PoW})}{P_t^{PoW}}$ . In contrast, the available supply of cryptoasset units in period  $t$  in a PoW blockchain is the total supply,  $M_t$ , minus those paid as fees,  $\frac{\int f_{(i,t-2)}^{PoS} dG(c_{(i,t-2)})}{P_t^{PoW}}$ , which accumulate to the users that mine and are therefore held until period  $t+1$ , minus those held by intermediately aged agents,  $\frac{G(c^{PoW})}{P_{t-1}^{PoW}}$ , who have no need to liquidate given that they derive no utility from consumption in that period.

PoS: In a PoS blockchain, all users that stake and invest in the DeFi option must purchase the PoS cryptoasset. Thus, the demand for a PoS cryptoasset in period  $t$  is given by the total number of coins that adopting users wish to purchase given by  $\frac{G(c^{PoS})}{P_t^{PoS}}$ . In contrast, the available supply of cryptoasset units in period  $t$  in a PoS blockchain is the total supply,  $M_t$ , minus those paid as fees,  $\frac{\int f_{(i,t-2)}^{PoS} dG(c_{(i,t-2)})}{P_t^{PoS}}$ , which accumulate to the users that stake and are therefore held until period  $t+1$ , minus those held by intermediately aged agents,  $\frac{G(c^{PoS})}{P_{t-1}^{PoS}}$ , who again have no need to liquidate until period  $t+1$ .

Therefore, using the fact that  $\frac{P_t^P}{P_{t-1}^P} = e^{-\rho}$  and equating the respective expressions and rearranging, we obtain that the PoW and PoS cryptoasset market clears whenever the following conditions hold:

$$\begin{aligned}\mathcal{M}^{PoW} &= (1 + e^{-\rho})(1 - \omega^{PoW})G(c^{PoW}) + \int f_{(i,t-2)}^{PoW} dG(c_{(i,t-2)}) \\ \mathcal{M}^{PoS} &= (1 + e^{-\rho})G(c^{PoS}) + \int f_{(i,t-2)}^{PoS} dG(c_{(i,t-2)})\end{aligned}\quad (25)$$

- (v) Validators are determined according to protocol rules In a PoW equilibrium, the total investment from users into mining computational power is equal to the total wealth invested into mining and DeFi activities,  $G(c^{PoW})$ , multiplied by the fraction of that wealth that is used for mining,  $\omega^{PoW}$ . Therefore,

$$H^*(1 + \kappa_h) = \omega^{PoW} \cdot G(c^{PoW}) \quad (26)$$

In a PoS equilibrium the amount of capital utilized in period  $t$  to purchase coins that will be staked in period  $t+1$  is equal to  $S^* = S_{t+1} P_t$ . Thus, using the fact that the amount of wealth invested in period  $t$  for the use of staking activities in period  $t+1$  is equal to the amount of wealth invested in period  $t$  into the PoS cryptoasset,  $G(c^{PoS})$ , multiplied by the fraction of wealth that is used for staking  $\omega^{PoS}$ . Thus, it must be the case that

$$S^* = \omega^{PoS} \cdot G(c^{PoS}) \quad (27)$$

- (vi) Blockchain survival probability varies according to protocol rules  
Given the attacker's problem, the probability that the blockchain survives through period  $t+1$  for any fixed realization of the attacker's benefit  $\Gamma_t$  is given by  $\pi_{t+1}^P = 1 - v_{t+1}^P(A_t^*(\Gamma_t))$ . Furthermore,

$$\pi_{t \rightarrow t+2}^P = \pi_t^P \cdot \pi_{t+1}^P \cdot \mathbb{E}_t[\pi_{t+2}^P], \quad (28)$$

where throughout, we assume for simplicity that  $\Gamma_t$  is public information at the beginning of period  $t$  so that the realization of  $\pi_t^P$  and  $\pi_{t+1}^P$  are known at the beginning of period  $t$ . In that case, we need only to consider the probability that the blockchain is attacked in period  $t+2$ , which in a PoW equilibrium, is given by

$$\mathbb{E}_t[\pi_{t+2}^{PoW}] = \mathbb{E}_t[1 - v_{t+2}^{PoW}(A_{t+1}^*(\Gamma_{t+1}))] = \mathbb{P}(\Gamma_{t+1} < H^* \cdot (1 + \kappa_h)) \quad (29)$$

Therefore, given that Agent  $(i, t)$  adopts the blockchain only when they know that the blockchain will not be successfully attacked in periods  $t$

and  $t+1$  (i.e., whenever  $\pi_t^{PoW} = \pi_{t+1}^{PoW} = 1$ ), then  $\Gamma_{t+1} \sim U[0, \bar{\Gamma}]$  for all  $t$ , combined with (29) yields:

$$\pi_{t \rightarrow t+2}^{PoW} = \pi^{PoW} \equiv \mathbb{P}(\Gamma_{t+1} < H^* \cdot (1 + \kappa_h)) = \min\left\{\frac{H^* \cdot (1 + \kappa_h)}{\bar{\Gamma}}, 1\right\} \quad (30)$$

Similarly, in a PoS equilibrium,

$$\mathbb{E}_t[\pi_{t+2}^{PoS}] = \mathbb{E}_t[1 - v_{t+2}^{PoS}(A_{t+1}^*(\Gamma_{t+1}))] = \mathbb{P}(\Gamma_{t+1} < \alpha \cdot S^*) \quad (31)$$

As with the PoW case, given that Agent  $(i, t)$  adopts the blockchain only when they know that the blockchain will not be successfully attacked in periods  $t$  and  $t+1$  (i.e., whenever  $\pi_t^{PoS} = \pi_{t+1}^{PoS} = 1$ ), then  $\Gamma_{t+1} \sim U[0, \bar{\Gamma}]$  combined with (31) yields:

$$\pi_{t \rightarrow t+2}^{PoS} = \pi^{PoS} \equiv \mathbb{P}(\Gamma_{t+1} < \alpha \cdot S^*) = \min\left\{\frac{\alpha \cdot S^*}{\bar{\Gamma}}, 1\right\} \quad (32)$$

## 2. Model Solution

We begin by solving for the optimal fees  $f_{(i,t)}^P$ :

### Lemma 1. Optimal Fees

Under any  $p \in \{PoW, PoS\}$  equilibrium the optimal fees  $f_{(i,t)}^P$  are given by:

$$\text{For all } (i, t): f_{(i,t)}^P = \begin{cases} \frac{1}{\Lambda} \int_0^{c_{(i,t)}} x dG(x) & \text{if } c_{(i,t)} < c^P \\ 0 & \text{if } c_{(i,t)} \geq c^P \end{cases} \quad (33)$$

**Lemma 33** arises because users with higher wait disutility experience a higher utility loss for a given wait time. As a consequence, in equilibrium, users with higher wait disutilities optimally pay higher fees and thereby experience lower wait times. More formally, **Lemma 33** demonstrates that, for all agents who adopt blockchain (i.e., agents such that  $c_{(i,t)} < c^P$ ), the equilibrium fee increases monotonically with user wait disutility:  $\frac{df_{(i,t)}^P}{dc_{(i,t)}} = \frac{c_{(i,t)}}{\Lambda} \frac{dG}{dx} \big|_{x=c_{(i,t)}} \geq 0$ . Furthermore, agents who prefer the outside option to adopting the blockchain (i.e., agents such that  $c_{(i,t)} \geq c^P$ ) optimally pay zero fees to blockchain validators because those agents do not transact via the blockchain and thus do not benefit from priority processing.

The remaining equilibrium solutions vary by protocol, so we discuss PoW and PoS separately in the remainder of this section. The following proposition provides the PoW equilibrium solution:

### Proposition 1. PoW Equilibrium

Any PoW equilibrium is characterized by an adoption cutoff,  $c^{PoW}$ , such that  $c_{(i,t)} < c^{PoW}$  if and only if  $U_{(i,t)}^{PoW} > 1 + \sigma$ . The PoW equilibrium is defined by

the market value  $\mathcal{M}^{PoW} := \mathcal{M}^{PoW}(c^{PoW})$ , the equilibrium return to mining  $R^{PoW} := R^{PoW}(c^{PoW})$ , the equilibrium fraction of wealth invested in mining  $\omega^{PoW} := \omega^{PoW}(c^{PoW})$ , the equilibrium hash rate  $H^* := H^*(c^{PoW})$ , and the blockchain survival probability  $\pi^{PoW} := \pi^{PoW}(c^{PoW})$  where the functions of the equilibrium variables (demonstrating their dependence on equilibrium adoption) are given explicitly as:

$$\omega^{PoW}(c^{PoW}) := \min\left\{\frac{(1-e^{-2\rho})G(c^{PoW}) + fees(c^{PoW})}{G(c^{PoW})(e^{-2\rho} \cdot \psi + 1 - \frac{1-\delta}{1+\kappa_h})}, 1\right\} \quad (34)$$

$$\mathcal{M}^{PoW}(c^{PoW}) := (1+e^{-\rho})(1-\omega^{PoW}(c^{PoW}))G(c^{PoW}) + fees(c^{PoW}) \quad (35)$$

$$\begin{aligned} H^*(c^{PoW})(1+\kappa_h) &:= \omega^{PoW}(c^{PoW}) \cdot G(c^{PoW}) \\ &= \min\left\{\frac{(1-e^{-2\rho})G(c^{PoW}) + fees(c^{PoW})}{e^{-2\rho} \cdot \psi + 1 - \frac{1-\delta}{1+\kappa_h}}, G(c^{PoW})\right\} \end{aligned} \quad (36)$$

$$R^{PoW}(c^{PoW}) := \frac{1-\delta}{1+\kappa_h} + \frac{(1-e^{-2\rho})(1-\omega^{PoW}(c^{PoW}))G(c^{PoW}) + fees(c^{PoW})}{H^*(c^{PoW})(1+\kappa_h)} \quad (37)$$

$$\pi^{PoW}(c^{PoW}) := \min\{\hat{\pi}^{PoW}(c^{PoW}), 1\} \quad (38)$$

where

$$\hat{\pi}^{PoW}(c^{PoW}) := \frac{1}{\Gamma} \cdot \min\left\{\frac{(1-e^{-2\rho})G(c^{PoW}) + fees(c^{PoW})}{e^{-2\rho} \cdot \psi + 1 - \frac{1-\delta}{1+\kappa_h}}, G(c^{PoW})\right\} \quad (39)$$

and

$$fees(c^{PoW}) := \frac{1}{\Lambda} \int_0^{c^{PoW}} \int_0^c x dG(x) dG(c) \quad (40)$$

Moreover, User  $(i, t)$ 's equilibrium expected utility from adopting the blockchain is given by  $U_{(i,t)}^{PoW} := U^{PoW}(c_{(i,t)}, c^{PoW})$ , where:

$$\begin{aligned} U^{PoW}(c_{(i,t)}, c^{PoW}) &:= \pi^{PoW}(c^{PoW}) \times (\Pi^{PoW}(c^{PoW}) \\ &\quad - \frac{1}{\Lambda} \int_0^{c_{(i,t)}} x dG(x) - \frac{c_{(i,t)}}{\Lambda} \times (G(c^{PoW}) - G(c_{(i,t)}))) \end{aligned} \quad (41)$$

and

$$\Pi^{PoW}(c^{PoW}) := \max\{e^{-2\rho}(1+\psi), \frac{1-\delta}{1+\kappa_h} + \frac{fees(c^{PoW})}{G(c^{PoW})}\}$$

Finally, the cutoff equilibrium is well-defined with the cutoff  $c^{PoW}$  given uniquely as follows:

$$c^{PoW} = \begin{cases} \sup\{c \geq 0 : U^{PoW}(c, c) \geq 1+\sigma\} & \text{if } \{c \geq 0 : U^{PoW}(c, c) \geq 1+\sigma\} \neq \emptyset \\ 0 & \text{otherwise} \end{cases} \quad (42)$$

Note that, per [Equations \(34\) - \(38\)](#), each equilibrium quantity can be written as a function of only one endogenous quantity, the equilibrium adoption cutoff  $c^{PoW}$ . Then, since the equilibrium adoption cutoff  $c^{PoW}$  is always well-defined, per [Equation \(42\)](#), the PoW equilibrium is also always well-defined.<sup>6</sup>

Next, we provide the solution for the PoS equilibrium:

### Proposition 2. PoS Equilibrium

Any PoS equilibrium is characterized by an adoption cutoff,  $c^{PoS}$ , such that  $c_{(i,t)} < c^{PoS}$  if and only if  $U_{(i,t)}^{PoS} > 1+\sigma$ . The PoS equilibrium is defined by the market value  $\mathcal{M}^{PoS} := \mathcal{M}^{PoS}(c^{PoS})$ , the equilibrium return to staking  $R^{PoS} := R^{PoS}(c^{PoS})$ , the equilibrium fraction of wealth invested in staking  $\omega^{PoS} := \omega^{PoS}(c^{PoS})$ , the equilibrium value of staked cryptoasset  $\mathcal{S}^* := \mathcal{S}^*(c^{PoS})$ , and the blockchain survival probability  $\pi^{PoS} := \pi^{PoS}(c^{PoS})$ , where the functions of the equilibrium variables (demonstrating their dependence on equilibrium adoption) are given explicitly as:

$$\omega^{PoS}(c^{PoS}) := \min\left\{\frac{(1-e^{-2\rho})G(c^{PoS}) + fees(c^{PoS})}{G(c^{PoS}) \cdot e^{-2\rho} \cdot \psi}, 1\right\} \quad (43)$$

$$\mathcal{M}^{PoS}(c^{PoS}) := (1+e^{-\rho})G(c^{PoS}) + fees(c^{PoS}) \quad (44)$$

$$\begin{aligned} \mathcal{S}^*(c^{PoS}) &:= \omega^{PoS}(c^{PoS}) \cdot G(c^{PoS}) \\ &= \min\left\{\frac{(1-e^{-2\rho})G(c^{PoS}) + fees(c^{PoS})}{e^{-2\rho} \cdot \psi}, G(c^{PoS})\right\} \end{aligned} \quad (45)$$

<sup>6</sup> As a technical aside, the PoW equilibrium is not necessarily unique, so we select  $c^{PoW}$  to correspond to the maximum adoption equilibrium whenever multiple equilibria exist. It is possible that no PoW equilibria with positive adoption exists (i.e.,  $U^{PoW}(c, c) < 0$  for all  $c \geq 0$ ); in this case, there is a unique (trivial) equilibrium, and we therefore select that unique equilibrium (i.e.,  $c^{PoW} = 0$ ) in that particular case. We apply the same equilibrium selection criteria for Proof-of-Stake (see footnote 7).

$$R^{PoS}(c^{PoS}) := e^{-2\rho} + \frac{(1 - e^{-2\rho})G(c^{PoS}) + fees(c^{PoS})}{S^*(c^{PoS})} \quad (46)$$

$$\pi^{PoS}(c^{PoS}) := \min\{\hat{\pi}^{PoS}(c^{PoS}), 1\} \quad (47)$$

where

$$\hat{\pi}^{PoS}(c^{PoS}) := \frac{\alpha}{\Gamma} \cdot \min\left\{\frac{(1 - e^{-2\rho})G(c^{PoS}) + fees(c^{PoS})}{e^{-2\rho} \cdot \psi}, G(c^{PoS})\right\} \quad (48)$$

and

$$fees(c^{PoS}) := \frac{1}{\Lambda} \int_0^{c^{PoS}} \int_0^c x dG(x) dG(c) \quad (49)$$

Moreover, User  $(i, t)$ 's equilibrium expected utility from adopting the blockchain is given by  $U_{(i,t)}^{PoS} := U^{PoS}(c_{(i,t)}, c^{PoS})$ , where

$$U^{PoS}(c_{(i,t)}, c^{PoS}) := \pi^{PoS}(c^{PoS}) \times (\Pi^{PoS}(c^{PoS}) - \frac{1}{\Lambda} \int_0^{c_{(i,t)}} x dG(x) - \frac{c_{(i,t)}}{\Lambda} \times (G(c^{PoS}) - G(c_{(i,t)}))) \quad (50)$$

and

$$\Pi^{PoS}(c^{PoS}) := \max\{e^{-2\rho}(1 + \psi), 1 + \frac{fees(c^{PoS})}{G(c^{PoS})}\}$$

Finally, the equilibrium adoption cutoff  $c^{PoS}$  is given uniquely as follows:

$$c^{PoS} = \begin{cases} \sup\{c \geq 0 : U^{PoS}(c, c) \geq 1 + \sigma\} & \text{if } \{c \geq 0 : U^{PoS}(c, c) \geq 1 + \sigma\} \neq \emptyset \\ 0 & \text{otherwise} \end{cases} \quad (51)$$

Note that, per [Equations \(43\) - \(47\)](#), each equilibrium quantity can be written as a function of only one endogenous quantity, the equilibrium adoption cutoff  $c^{PoS}$ . Then, since the equilibrium adoption cutoff  $c^{PoS}$  is always well-defined per [Equation \(51\)](#), the PoS equilibrium is also always well-defined.<sup>7</sup>

<sup>7</sup> As a technical aside, the PoS equilibrium is not necessarily unique, so we select  $c^{PoS}$  to correspond to the maximum adoption equilibrium whenever multiple equilibria exist. It is possible that no PoS equilibria with positive adoption exists (i.e.,  $U^{PoS}(c, c) < 0$  for all  $c \geq 0$ ); in this case, there is a unique (trivial) equilibrium, and we therefore select that unique equilibrium (i.e.,  $c^{PoS} = 0$ ) in that particular case. We apply the same equilibrium selection criteria for Proof-of-Work (see footnote 6).

### 3. Comparing PoW and PoS

We begin by comparing equilibrium security between a PoW blockchain and a PoS blockchain in general. Our primary insight from this analysis is that, while there exists a theoretical trade-off between PoW and PoS blockchains (see [Proposition 3](#)), PoS nonetheless generates higher equilibrium security than PoW under all realistic model parameters sourced using public data and prior empirical work in the academic literature (see [Corollary 1](#)).

#### 3.1 Comparison under general model parameters

Recall that a blockchain is secure when the cost of a successful attack is large enough to ensure that the attacker optimally chooses not to attack. Moreover, the cost of a successful attack is the cost of acquiring a large enough share of overall validation investment so that the attacker gains control over the blockchain's block production process. Blockchain security thus depends both on the equilibrium validation investment level (from nonattackers) and also the validation investment share needed to gain control of the blockchain (relative to nonattackers). Our analysis finds that a PoS blockchain generates a higher validation investment level than a PoW blockchain for all model parameters but nonetheless allows an attacker to gain control of the blockchain with a lower share of validation investment than a PoW blockchain. A PoS blockchain is thus more secure than a PoW blockchain if and only if its advantage with regard to higher equilibrium validation investment outweighs its disadvantage with regard to allowing the attacker to gain control with a lower share of validation investment. The following proposition formalizes this point, providing a general condition under which a PoS blockchain is more secure than a PoW blockchain and vice versa:

#### **Proposition 3.** Comparison of PoW and PoS Blockchain Security

Consider a PoW and a PoS blockchain with the same scale  $\Lambda$ . There exists a threshold,  $\zeta \in [0, 1]$ , such that the PoW blockchain is more (less) secure than the PoS blockchain when the relative capital needed to attack the PoS blockchain,  $\alpha$ , is below (above) the threshold. In particular:

$$\alpha \leq \zeta \Rightarrow \pi^{PoW} \geq \pi^{PoS} \quad \text{and} \quad \alpha \geq \zeta \Rightarrow \pi^{PoS} \geq \pi^{PoW} \quad (52)$$

The threshold,  $\zeta$ , is the ratio of PoW validation investment to PoS validation investment in the case whereby PoW and PoS generate identical total validator revenues. Explicitly, this ratio is given as follows:

$$\zeta := \frac{e^{-2\rho} \cdot \psi}{e^{-2\rho} \cdot \psi + 1 - \frac{1-\delta}{1+\kappa_h}} \quad (53)$$

Note that  $\zeta \leq 1$  for all model parameters where  $\zeta$  is defined as the ratio of PoW validation investment to PoS validation investment when PoW and PoS blockchains generate identical validator revenues. In turn, this fact

implies that, regardless of model parameters, PoS validation investment exceeds PoW validation investment whenever PoW and PoS generate identical revenues across all validators. This is the advantage for PoS relative to PoW discussed earlier which stems from the fact that PoS validation does not require expending the energy and depreciation costs of mining equipment. Nonetheless, as also discussed, PoS faces a countervailing disadvantage relative to PoW due to the design of many PoS blockchain protocols. To see that disadvantage, note that  $\alpha \leq 1$  holds for all model parameters, implying that, to succeed in an attack, an attacker needs only  $\alpha \leq 1$  capital for each unit of PoS validation investment whereas the attacker would instead require one unit of capital for each unit of PoW validation investment. Equation (52) clarifies how the two aforementioned effects trade off and determines whether PoW or PoS generates higher security in equilibrium.

Crucially, while Equation (52) covers all possible model parameters, not all model parameters are consistent with real-world conditions. In fact, in Section 3.3, we provide estimates for the set of realistic model parameters surveyed from public data sources and the academic literature. Moreover, we demonstrate that  $\alpha \geq \zeta$  always holds under that subset of model parameters, implying that PoS generates higher equilibrium security than PoW in practice.

### 3.2 A lower-dimensional condition

Given that our key point in Section 3.3 relies on calibrating our model to the real-world context, it is useful to first reduce the set of parameters that we need to estimate before demonstrating our point. More formally, in this section, we derive a sufficient condition that implies that PoS generates a higher equilibrium security than PoW. Notably, this sufficient condition involves only three model parameters and yet serves as a sufficient condition for any values of the other model parameters. Consequently, we are able to establish, in Section 3.3, that PoS generates higher equilibrium security than PoW while needing to estimate only three model parameters. We now proceed to state the lower-dimensional sufficient condition for PoS to generate higher equilibrium security than PoW:

**Proposition 4.** As in practice, suppose that the PoS blockchain requires a supermajority of no more than two-thirds for confirmation of blocks to the blockchain (i.e.,  $\eta \leq \frac{2}{3}$ ). Let  $\delta_{1Y}$  and  $\psi_{1Y}$  denote the annual rates for depreciation of mining equipment and DeFi returns respectively and suppose that  $\delta_{1Y} > \psi_{1Y}$ . Next, denote by  $\delta_T$  and  $\psi_T$  the depreciation and DeFi rates over any arbitrary horizon  $T$  determined as follows:

$$(1 + \delta_T) = (1 + \delta_{1Y})^T \quad (1 + \psi_T) = (1 + \psi_{1Y})^T$$

Then, for any horizon  $T$ , the following condition implies that  $\alpha \geq \zeta$  and thus that PoS generates higher equilibrium security than PoW:

$$\lambda \leq \frac{1}{2} \times \left(1 + \frac{\log(1 + \delta_T)}{\log(1 + \psi_T)}\right) \quad (54)$$

Crucially, the right-hand side of (54) is equivalent for any horizon  $T$  because

$$\frac{\log(1 + \delta_T)}{\log(1 + \psi_T)} = \frac{\log(1 + \delta_{1Y})}{\log(1 + \psi_{1Y})} \quad (55)$$

for all  $T$  and therefore the above inequality (54) is invariant to the horizon  $T$ .

Before proceeding to Section 3.3, we highlight four important points regarding Proposition 4 that are key to understanding the interpretation and robustness of Condition (54).

**Remark 1:** The hypothesis of  $\eta \leq \frac{2}{3}$  is not restrictive and is encoded directly into the protocol for all PoS blockchains (e.g., Casper FFG specification for Ethereum; see John et al. 2025). For more details, the interested reader may consult footnote 1 and references therein.

**Remark 2:** Note that our model can be interpreted as allowing both PoW and PoS attackers to borrow unlimited amounts of the numeraire to mount their attack.<sup>8</sup> In contrast,  $\lambda \geq 1$  specifically represents borrowing in the native cryptoasset, which is distinguished from borrowing in the numeraire in that it entails zero repayment after a successful attack. To clarify, a successful attack renders the cryptoasset value as zero. This implies that, after a successful attack, any borrowed native cryptoasset can be repaid without cost, whereas borrowed numeraire would still require repayment and therefore would impose a cost on the validator. Notably, we apply  $\lambda$  to only PoS blockchains in our model, implying that we allow the attacker to borrow cryptoasset to attack only a PoS blockchain and not a PoW blockchain. We employ this asymmetry, which disadvantages PoS, as a forward-looking matter. Yet, borrowing *any* cryptoasset at a sufficient scale for a successful attack is currently implausible for many reasons including, but not limited to, the overcollateralized nature of present day cryptoasset lending markets.<sup>9</sup> For that reason,  $\lambda \approx 1$  in practice. Nonetheless, to allow for the possibility of future developments in lending markets, we include  $\lambda$  particularly for PoS in the case that an attacker might be able to borrow the native cryptoasset from these lending markets and use that cryptoasset for staking. If we were to assume that it were possible for the PoW attacker to borrow the native PoW cryptoasset in mounting the PoW attack

<sup>8</sup> Namely, the deep-pocketed nature of the attacker and the objective (12) can be interpreted as each attacker having limited capital but access to borrowing in the numeraire with a (normalized) 0% interest rate.

<sup>9</sup> While it is possible to borrow native PoS cryptoassets like ETH in lending markets, the main lending markets all require these loans to be overcollateralized implying that it is not possible to use this form of leverage for validation purposes without posting alternative assets as collateral. Further, posting other assets as collateral for the purpose of mounting an attack is equivalent to borrowing numeraire (i.e., the case in which  $\lambda = 1$ ). Namely, unlike the native cryptoasset, any alternative asset posted as collateral will have value after a successful attack, forcing repayment by the attacker and thus producing the same attacker incentives as borrowing numeraire.

then  $\lambda - 1 \geq 0$  can be interpreted as the percentage of capital that a PoS attacker can borrow over and above the amount that a PoW attacker can borrow, given the same initial capital. In turn, within our model,  $\lambda = 1$  indicates that both the PoW and PoS attackers have the same (native cryptoasset) borrowing capability whereas  $\lambda > 1$  corresponds to a borrowing advantage for the PoS attacker. In light of this, in what follows, we refer to  $\lambda$  as the relative amount a PoS attacker can borrow in the native cryptoasset when compared to a PoW attacker.

**Remark 3:** Equation (54) is a sufficient condition for PoS to generate higher equilibrium security than PoW and not a necessary condition. More explicitly, the negation of Equation (54) does not imply that PoW generates higher equilibrium security than PoS. Notably, Equation (54) is derived by conservatively bounding mining energy costs by zero (i.e.,  $\kappa_h \geq 0$ ). In turn, explicitly allowing for positive energy costs (which represent a substantial cost of mining) would lead to a weaker condition than Equation (54). We opt to rely on the more stringent condition, Equation (54), because it entails fewer parameters to calibrate (thus demonstrating robustness to other model parameters including energy costs,  $\kappa_h$ ) and is sufficient to establish that PoS security exceeds PoW security in practice.

**Remark 4:** The condition of Equation (54) is horizon-invariant in the sense that if this condition holds for a fixed horizon  $T$  then the condition must also hold for all horizons (see Equation (55)). This latter property is helpful because it enables us to select an arbitrary horizon when examining the real-world context in Section 3.3. Given that empirical results in prior literature have typically been reported in annualized terms, we arbitrarily select an annual horizon in Section 3.3 without loss of generality per Proposition 4.

### 3.3 Parameter estimates

We will now show that Equation (54) holds under realistic assumptions regarding PoS versus PoW attacker ability to borrow the native cryptoasset to mount an attack (see Remark 2 above). In particular, in Appendix A, we source various estimates for PoW annualized mining hardware depreciation rates,  $\delta_{1Y}$ , and DeFi annualized returns,  $r_{1Y}$ , both as extrapolated from publicly available data as well as reported by various academic papers. These estimates vary depending on the source and, for DeFi yields, the particular application. Nonetheless, all estimates are consistent with PoS generating higher security than PoW.

Estimates for annual mining equipment depreciation rates are summarized in Table 1. Our main data source is theblock.co, which provides an index (starting in 2021) for the average price of an ASIC mining machine within three tiers of efficiency (see Appendix A for more details). Using that data we demonstrate that all efficiency tiers incur an average annual depreciation rate between 60.4% and 66.38%. In addition, Prat and Walter (2021) calibrate a model of Bitcoin mining, accounting for the depreciation of mining equipment due to

**Table 1**  
**Estimates for annual depreciation rates of ASIC mining equipment (see Appendix A)**

| Data source             | Estimate for average annual depreciation rate $\delta_{1Y}$ |
|-------------------------|-------------------------------------------------------------|
| theblock.co             | 60.4% - 66.38%                                              |
| Academic source         | Estimate for average annual depreciation rate $\delta_{1Y}$ |
| Prat and Walter 2021    | 47.7% - 61.3%                                               |
| De Vries and Stoll 2021 | 46.5% - 89%                                                 |
| Stinner 2021            | 61.45%                                                      |

the advancement of new technologies. Their calibration estimates generate an annual depreciation rate of 47.7%-61.3%. Similar estimates are obtained in De Vries and Stoll (2021) and Stinner (2021) and are summarized in Appendix A.

Next, we produce estimates for annual DeFi returns,  $\psi_{1Y}$ . Note that, within our model,  $\psi$  corresponds to the net return (in ETH) from an ETH investment through a DeFi application. For example,  $\psi$  may correspond to the net return from lending ETH at a blockchain lending platform. We specify the alternative investment as an ETH investment specifically because it represents the opportunity cost of staking and the opportunity cost of an investment, in general, pertains to the return of the best risk-equivalent alternative investment. To that end, note that staking ETH entails holding an unleveraged long spot position in ETH, akin to, for example, lending ETH at a lending platform.

We take two approaches to sourcing  $\psi_{1Y}$ . First, we observe that, absent frictions, the equilibrium return from staking ETH should match the return from any risk-equivalent alternative investment. In turn, we rely on ETH staking yields as our primary data source for determining  $\psi_{1Y}$ . Second, for robustness, we use ETH lending data from the largest lending platform, AAVE. In this case, we rely on both recent data from AAVE and from metrics reported by the academic literature. Both of these data sources imply our main conclusion that PoS is more secure than PoW in practice.

We compute rolling annualized returns on staked ETH using historical daily ETH staking return data sourced from *Compass Financial Technologies (FT)* where the staking return is calculated as the total daily rewards paid to validators as a percentage of staked ETH capital. This data set comports with the data provided by beaconcha.in, which is used by the Ethereum Foundation to report (annualized) daily staking returns (see <https://ethereum.org/en/staking/>).<sup>10</sup> Using the Compass FT daily return data, we construct the rolling annualized ETH staking returns from October 1, 2023 (the earliest date in the sample with 1 year of preceding daily return data), to the last date of the sample, June 10, 2024. We provide summary statistics for these rolling annualized returns in Table 2. As can be seen, the average staking return over the sample is 4.32% with a minimum of 3.72% and maximum of 4.85%.

<sup>10</sup> We use Compass FT data because historical beaconcha.in data is not publically available.

**Table 2**  
Summary statistics of rolling annual staked ETH returns using Compass FT data

| Variable                        | Average | Median | Min   | Max   | SD    |
|---------------------------------|---------|--------|-------|-------|-------|
| Rolling staked ETH return (Ann) | 4.32%   | 4.33%  | 3.73% | 4.85% | .0031 |

As can be seen from [Tables 1](#) and [2](#), we develop a range of realistic parameter values for  $\delta_{1Y}$  of [46.5%, 89%] and a range of realistic parameter values for  $\psi_{1Y}$  of [3.73%, 4.85%]. In addition, for all estimates the condition  $\delta_{1Y} > \psi_{1Y}$  (equivalently  $\delta_T > \psi_T$  for all  $T$ ) is satisfied and therefore the sufficient condition of [Proposition 4](#) applies. Applying even the most conservative of these estimates,  $\delta_{1Y}=46.5\%$  and  $\psi_{1Y}=4.85\%$  to [Equation \(54\)](#) thus suggests that PoS is more secure than PoW so long as  $\lambda \leq 453\%$ . Therefore, PoS is more secure than PoW so long as the PoS attacker is unable to borrow more than 353% more native cryptoasset capital than a PoW attacker to mount an attack of the blockchain. In a similar vein, using less conservative estimates of  $\delta_{1Y}=55\%$  and  $\psi_{1Y}=4.32\%$  then PoS is more secure than PoW whenever  $\lambda \leq 568\%$  or equivalently whenever the PoS attacker is unable to borrow more than 468% more native cryptoasset capital than a PoW attacker to mount an attack. We summarize these findings via the next corollary:

**Corollary 1.** Suppose that the PoS blockchain requires a super-majority of no more than two-thirds (i.e.,  $\eta \leq \frac{2}{3}$ ). Suppose also that, per [Table 1](#), the mining hardware depreciation rate,  $\delta_{1Y}$ , is at least 46.5% whereas, per [Table 2](#), the return on DeFi investment,  $\psi_{1Y}$ , is no more than 4.85%. Then,  $\alpha \geq \zeta$  and therefore PoS generates higher equilibrium security than PoW whenever  $\lambda \leq 453\%$ . Using less conservative estimates whereby  $\delta_{1Y}$  is at least 55% and  $\psi_{1Y}$  is no more than 4.32% then implies that PoS generates higher equilibrium security than PoW whenever  $\lambda \leq 568\%$ .

[Corollary 1](#) implies that a PoS attacker would need to be able to (unrealistically) borrow substantially more of the native cryptoasset than a PoW attacker (353% and 468% more, resp.) in order for a PoW blockchain to generate higher security than a PoS blockchain under the same real-world conditions. Importantly, borrowing in cryptoasset markets currently remains largely (over) collateralized, requiring custody of cryptoasset collateral. Thus, it is not possible to use these platforms to acquire leverage for validation purposes in the context of our model (see footnote 10). Furthermore, we note that, to date, there is no differentiating factor in PoS versus PoW lending market structure that would suggest any greater ability to borrow PoS versus PoW native cryptoasset for validation purposes (i.e., all platforms lend both native cryptoasset types). This suggests that  $\lambda \approx 1$  in practice and therefore that PoS is more secure than PoW in practice.

Finally, as mentioned above, we also employ data for ETH lending to the largest blockchain lending platform, AAVE, for robustness. To that end, To

**Table 3**  
Annualized DeFi ETH lending return,  $\psi$  (see [Appendix A](#))

| Source                                                        | Annualized DeFi ETH lending return $\psi$ |
|---------------------------------------------------------------|-------------------------------------------|
| AAVE                                                          | 1.66% average (2023-2024)                 |
| <a href="#">Chaudhary, Kozhan, and Vishwanath-Natraj 2023</a> | 0.13% average (max 2.28%)                 |

that end, we also report in [Table 3](#) (and [Appendix A](#)) some additional sources for historical DeFi returns, such as the AAVE 1-year average ETH lending return and other DeFi lending returns, as reported by the academic literature. As can be seen, these additional estimates do not affect the relevant parameter bounds of  $\psi$  and therefore the main conclusions regarding attacker leverage (as summarized in [Corollary 1](#)) are unchanged under these estimates. Moreover, for further robustness, we report alternative asset lending rates in [Appendix A](#) and demonstrate that our parameter bounds are still unaffected by these additional estimates for average DeFi rates of return.

#### 4. Scaling Results

We now turn to examining high-scale blockchains (i.e., blockchains with large  $\Lambda$ ) while focusing on practical insights. To that end, we restrict ourselves to realistic parameters per the findings of [Section 3.3](#) (i.e.,  $\alpha \geq \zeta$ ). Our primary insight herein is that the security advantage for PoS blockchains relative to PoW blockchains is particularly salient at high-scale. In particular, given a large maximal attacker benefit (i.e., large  $\bar{\Gamma}$ ), scaling can undermine blockchain security, leading to full insecurity for a blockchain (i.e.,  $\pi^P=0$ ). Crucially, via [Proposition 5](#), we show that this security deterioration is more urgent for a PoW blockchain than for a PoS blockchain in that it occurs for a larger set of maximal attacker benefits for a PoW blockchain relative to a PoS blockchain. As a contrast, given a small maximal attacker benefit (i.e., small  $\bar{\Gamma}$ ), scaling can enhance security for a blockchain, leading to full security for a blockchain (i.e.,  $\pi^P=1$ ). We show, via [Proposition 7](#), that this security enhancement is more relevant for a PoS blockchain than for a PoW blockchain in that it occurs for a larger set of maximal attacker benefits for a PoS blockchain relative to a PoW blockchain. Intuitively, scaling can have positive or negative effects (depending on the maximal attacker benefit), yet the negative effects impact PoW blockchains before impacting PoS blockchains, while the positive effects impact PoS blockchains before impacting PoW blockchains.

##### 4.1 Scaling security risks are more urgent for PoW relative to PoS

As a preliminary step for our analysis, it is useful to define a set,  $\Sigma_0^P$ , consisting of all maximal attacker benefits,  $\bar{\Gamma}$ , which generate full insecurity (i.e.,  $\pi^P=0$ ) for high-scale blockchains (i.e., for sufficiently large  $\Lambda$ ):

$$\Sigma_0^p := \{\bar{\Gamma} > 0 : \exists \underline{\Lambda}^p > 0 \text{ s.t. } \pi^p = 0 \text{ for all } \Lambda > \underline{\Lambda}^p\} \quad (56)$$

where  $p \in \{PoW, PoS\}$  denotes the blockchain's protocol. With this definition established, we now state our first result regarding high-scale blockchains:

**Proposition 5.** High-scale PoW blockchains are insecure for larger attacker benefit set

Per [Section \(3.3\)](#), assume that  $\alpha \geq \zeta$ . Then, the set of maximal attacker benefits that generates full insecurity for high-scale PoW blockchains is larger than the set of maximal attacker benefits that generates full insecurity for high-scale PoS blockchains:

$$\Sigma_0^{PoS} \subseteq \Sigma_0^{PoW} \quad (57)$$

In turn, scaling can render a PoW blockchain fully insecure in cases when it does not render a PoS blockchain fully insecure. Moreover, scaling cannot render a PoS blockchain fully insecure unless it also renders a PoW blockchain fully insecure.

[Proposition 5](#) establishes that the potential security risk from scaling a blockchain is more pressing for a PoW blockchain than for a PoS blockchain. In particular, scaling can render a blockchain fully insecure when the attacker's maximal benefit is sufficiently large, and [Proposition 5](#) establishes that this effect arises for a larger set of maximal attacker benefits for a PoW blockchain than for a PoS blockchain. More formally, [Proposition 5](#) establishes  $\Sigma_0^{PoS} \subseteq \Sigma_0^{PoW}$ , which implies that there exists a set of maximal attacker benefits under which scaling renders a PoW blockchain fully insecure but does not render a PoS blockchain fully insecure. Notably, [Proposition 5](#) also implies that there does not exist *any* maximal attacker benefit under which scaling would render a PoS blockchain fully insecure while not also rendering a PoW blockchain fully insecure. [Proposition 5](#) thus establishes that the security risk of scaling apply to PoW in cases in which it does not apply to PoS, but not vice versa, thereby highlighting that scaling generates a more pressing risk for PoW than for PoS.

To expand further on the aforementioned insight, the sense in which scaling poses a higher security risk for PoW than for PoS can also be seen from a different vantage point. More explicitly, if we fix a maximal attacker benefit level,  $\bar{\Gamma}$ , then we can examine when scaling becomes problematic for PoW relative to PoS. As we will show, scaling generates security failure for PoW at lower scale than for PoS. More formally, we define the supremum of scale,  $\Lambda > 0$ , such that a blockchain with protocol  $p \in \{PoW, PoS\}$  maintains positive security,  $\bar{\Lambda}_0^p$ , as follows:

$$\bar{\Lambda}_0^p = \sup\{\Lambda > 0 : \pi^p > 0\}, \quad (58)$$

In turn, we provide the following result, which clarifies that scaling becomes problematic for PoW at lower scale than for PoS:

**Proposition 6.** PoW blockchains become fully insecure at lower scale than PoS blockchains

Per Section (3.3), assume that  $\alpha \geq \zeta$  and fix the attacker's maximal benefit as  $\bar{\Gamma} > 0$ . Then, the level of scale whereby the blockchain maintains a positive level of security (i.e.,  $\pi^p > 0$ ) is higher for PoS than for PoW:

$$\bar{\Lambda}_0^{PoS} \geq \bar{\Lambda}_0^{PoW} \quad (59)$$

In turn, there exist a range of scale,  $(\bar{\Lambda}_0^{PoW}, \bar{\Lambda}_0^{PoS})$ , such that scaling the blockchain within that range renders a PoW blockchain fully insecure but does not render a PoS blockchain fully insecure.

As a technical aside, when  $\bar{\Gamma} \notin \Sigma_0^{PoW}$ , then Proposition 5 implies  $\bar{\Gamma} \notin \Sigma_0^{PoS}$ . In that case, the definitions of  $\Sigma_0^{PoS}$  and  $\Sigma_0^{PoW}$  imply that  $\bar{\Lambda}_0^{PoS} = \bar{\Lambda}_0^{PoW} = \infty$ , yielding that Proposition 6 holds trivially (i.e.,  $\infty \leq \infty$ ). Additionally, when  $\bar{\Gamma} \notin \Sigma_0^{PoS}$  but  $\bar{\Gamma} \in \Sigma_0^{PoW}$ , then  $\bar{\Lambda}_0^{PoS} = \infty$  but  $\bar{\Lambda}_0^{PoW} < \infty$  in which case Proposition 6 follows immediately (i.e.,  $\bar{\Lambda}_0^{PoW} < \bar{\Lambda}_0^{PoS} = \infty$ ). Consequently, the only nontrivial case is  $\bar{\Gamma} \in \Sigma_0^{PoS} \subseteq \Sigma_0^{PoW}$  where the definitions of  $\Sigma_0^{PoW}$  and  $\Sigma_0^{PoS}$  imply  $\bar{\Lambda}_0^{PoS}, \bar{\Lambda}_0^{PoW} < \infty$ . In that case, Proposition 6 establishes that scaling erodes PoW security at a lower finite scale than it erodes PoS security (i.e.,  $\bar{\Lambda}_0^{PoW} \leq \bar{\Lambda}_0^{PoS} < \infty$ ). Notably, there exist a range of scale,  $(\bar{\Lambda}_0^{PoW}, \bar{\Lambda}_0^{PoS})$ , such that PoW is fully insecure while PoS maintains some level of security.

Before proceeding to our next set of results, we offer some intuition regarding why scaling can undermine security per both Propositions 5 and 6. We first note that blockchain user fees are paid to validators and thus serve as an incentive for validation investment. Scaling reduces equilibrium user fees and can thereby reduce validator revenue, which then reduces the incentive for validation investment. In turn, lower validation investment reduces security because it reduces the cost of an attack. To be more explicit, recall from Equation (33) that the equilibrium fee,  $f_{(i,t)}^p$ , paid by blockchain user  $(i, t)$  is given as follows:

$$f_{(i,t)}^p = \frac{1}{\Lambda} \int_0^{c(i,t)} x \, dG(x), \quad p \in \{PoW, PoS\}. \quad (60)$$

Equation (60) directly implies that a blockchain user's equilibrium fee decreases with the blockchain's scale (i.e.,  $\frac{df_{(i,t)}^p}{d\Lambda} < 0$ ) and goes to zero as the blockchain scale diverges (i.e.,  $\lim_{\Lambda \rightarrow +\infty} f_{(i,t)}^p = 0$ ). Thus, scaling a blockchain reduces user fees and therefore can also reduce revenues for validators. Importantly, when the revenue for validating decreases then so too does the equilibrium investment into validation activities. Moreover, a decrease in validation investment implies an increase in the probability of successful

attack. Of particular note, when the maximal attacker benefit is sufficiently large, then scaling can generate a sufficient reduction in validation investment such that the resulting security level is too low to maintain *any* user adoption. In that case, scaling the blockchain will generate both zero adoption and zero security.

To understand the last part of the aforementioned argument regarding zero user adoption, it is important to recall that agents have access to an outside option that generates a return of  $1 + \sigma > 1$ . Thus, in order for the blockchain to generate *any* level of adoption, it must be the case that blockchain adoption generates a return larger than  $1 + \sigma$  for some agents. Crucially, when the maximal attacker benefit is large, then scaling can reduce user fees and thereby validator revenues and validation investment to such an extent that the return from adopting blockchain is necessarily below  $1 + \sigma$ . This effect arises specifically due to the security reduction from the lower validation investment in the context of a high maximal attacker benefit. To provide some technical context, we note that the supremum of user returns can be bounded above as follows:<sup>11</sup>

$$\sup_{(i,t)} U_{(i,t)}^p \leq \pi^p \cdot \chi, \quad p \in \{PoW, PoS\} \quad (61)$$

where  $\chi := \max\{1 + \psi, 1 + \frac{1}{\Lambda} \int_0^\infty x \, dG(x)\} < \infty$  is a finite upper bound for the return from adopting the blockchain if the blockchain were fully secure. As a consequence, if  $\pi^p$  falls sufficiently then all user returns necessarily fall below the outside option return. More explicitly:

$$\pi^p < \frac{1 + \sigma}{\chi} \Rightarrow \sup_{(i,t)} U_{(i,t)}^p < 1 + \sigma \quad (62)$$

In turn,  $\sup_{(i,t)} U_{(i,t)}^p < 1 + \sigma$  implies zero adoption (i.e.,  $c^p = 0$ ). Crucially, zero adoption implies zero fee revenue and also zero block reward revenue because zero adoption implies zero demand for the cryptoasset and thus zero value for any inflationary rewards. Hence, zero revenue for validators implies zero validation investment and thus zero security (i.e.,  $\pi^p = 0$ ). As such, when the maximal attacker benefit is sufficiently high, scaling can reduce fees to such an extent that the resulting security loss is incompatible with any user adoption, implying that the scaling generates zero adoption and thereby zero security.

<sup>11</sup> More explicitly, Equation (41) implies that  $U_{(i,t)}^{PoW} \leq \pi^{PoW} (e^{-2\rho(1+\psi)}) \leq \pi^{PoW} (1 + \psi)$ . Moreover, Equation (50) implies that  $U_{(i,t)}^{PoS} \leq \pi^{PoS} \max\{e^{-2\rho(1+\psi)}, 1 + \frac{fees(c^{PoS})}{G(c^{PoS})}\} \leq \pi^{PoS} \max\{1 + \psi, 1 + \frac{1}{\Lambda} \int_0^{c^{PoS}} x \, dG(x)\} \leq \pi^{PoS} \max\{1 + \psi, 1 + \frac{1}{\Lambda} \int_0^\infty x \, dG(x)\}$  where the middle inequality comes from the fact that  $fees(c^{PoS}) = \int_0^{c^{PoS}} \int_0^c x \, dG(x) \, dG(c) \leq \int_0^{c^{PoS}} \int_0^{c^{PoS}} x \, dG(x) \, dG(c) = G(c^{PoS}) \int_0^{c^{PoS}} x \, dG(x)$ . Generalizing across the two cases, we have  $U_{(i,t)}^p \leq \pi^p \cdot \chi$  where  $\chi := \max\{1 + \psi, 1 + \frac{1}{\Lambda} \int_0^\infty x \, dG(x)\}$ , and this directly implies Equation (61).

Notably, the notion of the maximal attacker benefit being sufficiently high for scaling to render a blockchain fully insecure depends on the protocol. In particular, our results highlight that this notion encompasses a larger set of maximal attacker benefits for PoW than for PoS, yielding that the negative effects of scaling are more pressing for PoW than for PoS. More formally, [Proposition 5](#) demonstrates that scaling renders PoW fully insecure for a larger range of maximal attacker benefits than PoS.

#### 4.2 Scaling security advantages are more immediate for PoS relative to PoW

Prior to stating our next set of results, we define a set,  $\Sigma_1^p$ , consisting of all levels of maximal attacker benefit,  $\bar{\Gamma}$ , that generate full security (i.e.,  $\pi^p = 1$ ) for high-scale blockchains (i.e., for sufficiently large  $\Lambda$ ):

$$\Sigma_1^p := \{\bar{\Gamma} > 0 : \exists \underline{\Lambda}^p > 0 : \forall \Lambda > \underline{\Lambda}^p : \pi^p = 1\} \quad (63)$$

where  $p \in \{PoW, PoS\}$  denotes the blockchain's protocol. With the definition of  $\Sigma_1^p$  established, we now state our next result regarding high-scale blockchains:

**Proposition 7.** High-scale PoS blockchains are fully secure for larger attacker benefit set

Per [Section \(3.3\)](#), assume that  $\alpha \geq \zeta$ . Then, the set of maximum attacker benefits that generates full security for high-scale PoS blockchains is larger than the set of maximum attacker benefits that generates full security for high-scale PoW blockchains:

$$\Sigma_1^{PoW} \subseteq \Sigma_1^{PoS} \quad (64)$$

Thus, scaling enables a PoS blockchain to achieve fully security in cases that do not generate full security for a PoW blockchain. Moreover, scaling cannot generate full security for a PoW blockchain if it does not also generate full security for a PoS blockchain.

[Proposition 7](#) establishes that the potential benefit from scaling is more relevant for a PoS blockchain than for a PoW blockchain. More explicitly, scaling can generate full security for a blockchain when the maximal attacker benefit is sufficiently low; [Proposition 7](#) shows that this effect arises for a larger set of maximal attacker benefits for a PoS blockchain than for a PoW blockchain. More formally, [Proposition 7](#) demonstrates that  $\Sigma_1^{PoW} \subseteq \Sigma_1^{PoS}$ , implying that there exists a set of maximal attacker benefits under which scaling induces full security for a PoS blockchain, but not for a PoW blockchain. Notably, [Proposition 7](#) also implies that there exist no conditions such that scaling generates full security for a PoW blockchain, but not for a PoS blockchain. Rather, if scaling generates full security for a PoW blockchain, then it necessarily also generates full security for a PoS blockchain. [Proposition 7](#) thus establishes that the benefit of scaling applies to PoS in cases in which it

does not apply to PoW, but not vice versa. Intuitively, the benefit of scaling is more directly relevant for PoS than for PoW.

To expand further, the sense in which the benefit of scaling is more relevant for PoS than for PoW can also be seen through a different lens. More explicitly, if we fix a maximal attacker benefit level,  $\bar{\Gamma}$ , then we can examine when scaling generates full security for PoS relative to PoW. As we will show, scaling generates full security for PoS at a lower scale than for PoW. To see this point, as a preliminary, we define the supremum of scale,  $\Lambda > 0$ , such that a blockchain with protocol  $p \in \{PoW, PoS\}$  does not possess full security,  $\bar{\Lambda}_1^p$ , as follows:

$$\bar{\Lambda}_1^p = \sup\{\Lambda > 0 : \pi^p < 1\} \quad (65)$$

Crucially,  $\bar{\Lambda}_1^p$  denotes the scale at which the blockchain achieves full security whenever  $\Lambda > \bar{\Lambda}_1^p$ . This is because any scale above  $\bar{\Lambda}_1^p$  implies full security (i.e.,  $\Lambda > \bar{\Lambda}_1^p \Rightarrow \pi^p = 1$ ) by the definition from Equation (65). With that context given, we offer the following result:

**Proposition 8.** PoS blockchains become fully secure at lower scale than PoW blockchains

Per Section (3.3), assume that  $\alpha \geq \zeta$  and fix the attacker's maximal benefit as  $\bar{\Gamma}$ . Then, the scale whereby the blockchain achieves full security (i.e.,  $\pi^p = 1$ ) is lower for PoS than for PoW:

$$\bar{\Lambda}_1^{PoS} \leq \bar{\Lambda}_1^{PoW} \quad (66)$$

In turn, there exist a range of scale,  $(\bar{\Lambda}_1^{PoS}, \bar{\Lambda}_1^{PoW})$ , such that scaling the blockchain within that range enables a PoS blockchain to achieve full security but a PoW blockchain does not similarly achieve full security.

To provide some technical background, when  $\bar{\Gamma} \notin \Sigma_1^{PoS}$ , then Proposition 7 implies  $\bar{\Gamma} \notin \Sigma_1^{PoW}$ . In that case, the definitions of  $\Sigma_1^{PoS}$  and  $\Sigma_1^{PoW}$  imply  $\bar{\Lambda}_1^{PoS} = \bar{\Lambda}_1^{PoW} = \infty$ , yielding that Proposition 8 holds trivially when  $\bar{\Gamma} \notin \Sigma_1^{PoS}$  and  $\bar{\Gamma} \notin \Sigma_1^{PoW}$ . In the alternative case that  $\bar{\Gamma} \in \Sigma_1^{PoS}$  but  $\bar{\Gamma} \notin \Sigma_1^{PoW}$ , then  $\bar{\Lambda}_1^{PoS} < \infty$  but  $\bar{\Lambda}_1^{PoW} = \infty$ . In turn, in that case, Proposition 8 follows immediately from  $\bar{\Lambda}_0^{PoS} < \infty = \bar{\Lambda}_0^{PoW}$ . Thus, the only nontrivial case is  $\bar{\Gamma} \in \Sigma_1^{PoS} \subseteq \Sigma_1^{PoW}$  where the definitions of  $\Sigma_1^{PoS}$  and  $\Sigma_1^{PoW}$  imply  $\bar{\Lambda}_1^{PoS}, \bar{\Lambda}_1^{PoW} < \infty$ . In that case, Proposition 8 establishes that scaling enables PoS to achieve full security at a lower finite scale than PoW (i.e.,  $\bar{\Lambda}_1^{PoS} \leq \bar{\Lambda}_1^{PoW} < \infty$ ). Notably, there exist a range of finite scales,  $(\bar{\Lambda}_1^{PoS}, \bar{\Lambda}_1^{PoW})$ , such that PoS is fully secure but PoW does not achieve full security.

To provide some context regarding why scaling can generate full security, we note that scaling can adoption, thus enabling full security. In particular, as discussed, scaling alleviates congestion, which reduces equilibrium user fees. While the fee reduction has the partial effect of reducing revenues to

validators, the fee reduction also increases user blockchain demand because user utility is decreasing in fees. Crucially, when the attacker's maximal benefit is sufficiently low, the partial effect of lower fees to validators is offset by increased blockchain adoption due to lower fees. In more detail, increased blockchain adoption implies increased demand for the native cryptoasset, which then implies a higher market value for the native cryptoasset and thereby a higher value for block rewards paid to validators. When the maximal attacker benefit is sufficiently low, then the increase in block reward value offsets the validator revenue loss due to lower fees and thus an increase in scale generates both higher adoption and higher security.

Crucially, the notion of the maximal attacker benefit being sufficiently low for a blockchain to benefit from scaling depends on the protocol. Our results highlight that the notion of a sufficiently low attacker benefit is broader for PoS than for PoW, yielding that PoS is more likely to benefit from scaling. More formally, [Proposition 7](#) demonstrates that PoS experiences the potential benefit from scaling (i.e., full security) for a larger range of maximal attacker benefits than PoW.

## 5. Conclusion

In this paper we study the equilibrium relationship among blockchain scale, security, and adoption. Of particular note, we derive a condition to demonstrate the trade-off between PoW and PoS blockchain security. Notably, we clarify that, under current real-world conditions, PoS blockchains are likely to be more secure than PoW blockchains.

It is important to understand that our work is fundamental to the modern blockchain literature that focuses on tokenomics (see, e.g., [Cong, Li, and Wang 2021](#); [Gan, Tsoukalas, and Netessine 2021a](#); [Gan, Tsoukalas, and Netessine 2021b](#); [Goldstein, Gupta, and Sverchkov 2023](#); [Mayer 2022](#)) and Decentralized Finance (see, e.g., [Lehar and Parlour 2025](#); [Capponi and Jia 2022](#); [Hasbrouck, Rivera, and Saleh 2022](#); [John, Kogan, and Saleh 2023](#)). In more detail, the ability for smart contracts and decentralized applications to operate as specified presumes that the blockchain is not disrupted. In turn, our paper provides a foundational economic framework to understand whether the welfare effects studied by the tokenomics and decentralized finance literature can actually be realized.

## Appendix A: Model Parameter Estimates

In this appendix we will provide various estimates for the key model parameters; the mining depreciation rate  $\delta$  and the DeFi return  $\psi$ .

### A.1. Estimates for the Depreciation of Mining Equipment ( $\delta$ )

In what follows we list a number of estimates for the annual depreciation of mining equipment coming from both the academic literature as well as using public data on the price of mining equipment.

- **Prat and Walter (2021)** study a model of PoW mining and estimate the rate of technological progress, which they denote by  $a$ . In particular, the value  $a$  is such that cost of investment,  $I_1$  to produce 1 unit of hash power 1 year from now evolves as  $I_1 = e^{-a} \cdot I_0$  where  $I_0$  is the cost of producing 1 unit of hash power today. Thus, the value of existing mining equipment is declining with technological progress  $a$ . Assuming a competitive mining market whereby the value of the equipment is equal to the discounted value of future revenues, this implies a rate of technological/hardware depreciation equal to  $\frac{I_0 - I_1}{I_0} = 1 - e^{-a}$ . The authors then perform a calibration exercise and estimate the annual value of  $a$  from 2018 to 2020 as ranging between .65 and .95. These estimates therefore generate an implied annual depreciation rate between 47.7% and 61.3%.
- **De Vries and Stoll (2021)** study the lifetime of ASIC mining devices. They estimate that the average profitable lifetime of such a device is between 1.12 and 2.15 years from July 2016 until July 2020. These estimates generate an implied annual depreciation rate between 46.5% and 89%.
- **Stinner (2021)** simulates a model of bitcoin mining and estimates the average useful lifespan of a mining machine to be 594 days, generating an implied annual depreciation rate of 61.45%.
- We further produce similar estimates as the above sources, using data from the block.co (<https://www.theblock.co/data/on-chain-metrics/bitcoin/bitcoin-asic-price-index>), which documents three ASIC price indices consisting of the average price per unit in three efficiency categories: <25 J/TH, 25-38 J/TH, and >38 J/TH.<sup>12</sup> As documented in Table A1, the ASIC price index across each category drops on average by no less than 60% per year over the first 2 years of the data sample (August 2021-2023) implying an annual depreciation rate between 60.38% and 60.92%.
- Corroborating the idea that these depreciation estimates are consistent with long term estimates, we note that Bitmain (one of the largest producers of ASIC Bitcoin mining equipment) listed in its IPO registration statement (form F1) that one of the company's main risk factors stemmed from the fact that the average selling prices will decrease over time due to technological advancement. The registration statement is available at [https://www.sec.gov/Archives/edgar/data/1780652/000119312519276263/d773846df1.htm#from773846\\_3](https://www.sec.gov/Archives/edgar/data/1780652/000119312519276263/d773846df1.htm#from773846_3). We reproduce the relevant text directly from the registration statement here:

**“The average selling prices of our products may decrease from time to time due to technological advancement and we may not be able to pass onto our suppliers such decreases, which may in turn adversely affect our profitability.**

<sup>12</sup> The block.co reports that the ASIC price index is calculated as follows: “The ASIC Price Index reflects the current price per TH of different Bitcoin mining ASICs grouped by three efficiency tiers. Luxor calculates the ASIC Price Index by averaging each unit price based on its nominal hashrate output. They group together different ASIC models into different efficiency tiers based on their power efficiency and display each efficiency in terms of \$ per TH. Each Bitcoin mining ASIC listing they use in the index is vetted through a confidence interval using previous index values and our proprietary algorithm.

The three efficiency tiers:

- Under 38 joules/TH (38 J/TH) includes the latest generation Bitcoin mining ASICs like Bitmain's Antminer S19, S19j and S19 Pro units and MicroBTC's Whatsminer M30s, M30s+ and M30s++ models, among others.
- 38-to-68 J/TH includes mid-generation ASICs like Bitmain's AntminerS17 and T17 series and MicroBT's Whatsminer M20, M21 and M32 series models, among others.
- Over 68 J/TH includes old-generation ASICs like Bitmain's Antminer S9, T9, S11, and T15 series and MicroBT's Whatsminer M10 series, among others.”

**Table A1**  
**Average Price Index change YoY from 2021 to 2023 by efficiency tier**

| Efficiency tier | Price change<br>'21-'22 | Price change<br>'22-'23 | Price change per year avg.<br>'21-'23 |
|-----------------|-------------------------|-------------------------|---------------------------------------|
| < 25 J/TH       | -61.36%                 | -60.47%                 | -60.92%                               |
| 25-38 J/TH      | -61.04%                 | -59.77%                 | -60.4%                                |
| 38 J/TH         | -67.53%                 | -65.22%                 | -66.38%                               |

The IC design industry is characterized by rapid launches of new products, continuous technological advancements and changing market trends and customer preferences, all of which translate to a shorter life cycle and a gradual decrease in the average selling prices of products over time. For example, the average selling price per Thash for our Bitcoin mining machines decreased from RMB613 in 2017 to RMB369 (US\$54) in 2018 and further decreased to RMB72 (US\$10) in the 6 months ended June 30, 2019. Because we compete in the environment of rapidly evolving technology advancement and market trends and developments of the IC design industry, there are no assurances that we will be able to pass on any decrease in average selling prices of our products to our suppliers. In the event that average selling prices of our products unusually or significantly decrease and such decreases cannot be offset by a corresponding decrease in the prices of the principal components of our products, our gross profit margins may be materially and adversely affected, which in turn, may adversely affect our profitability.”

**A.2 Estimates for DeFi Yields ( $\psi$ )**

In this subsection we will demonstrate various estimates for DeFi yields, also taken from the academic literature and publicly available data.

- We start with information about ETH staking yields. Note that in equilibrium these yields should approximate DeFi yields from investments that bear similar risk to staking as otherwise some DeFi investors will optimally switch from DeFi investing to staking (and vice versa). We use publicly available data from Compass Financial Technologies (<https://www.compassft.com/indice/styeth/>), which reports their methodology here: [https://www.compassft.com/wp-content/uploads/Methodology\\_Compass\\_Staking\\_Yield\\_Reference\\_Index.pdf](https://www.compassft.com/wp-content/uploads/Methodology_Compass_Staking_Yield_Reference_Index.pdf). We note that this data comports with the open source data provided by Beaconcha.in <https://beaconcha.in/ethstore>, a staking return tracker that represents the code that reports daily staking yields on Ethereum’s official website <https://ethereum.org/en/staking/> (Compass makes their data publicly available, so we use this source instead of Beaconcha.in). Next, using the daily return data from Compass Financial Technologies we calculate the rolling annual return from staking from October 1, 2023 (the earliest date with 1 year of preceding daily return data), to June 10, 2024 (the last date of the sample). We report summary statistics regarding the annual rolling staking returns in Table A2. As can be seen, the average staking return has been 4.32% over the 2-year sample, with a minimum of 3.73% and a maximum of 4.85%.
- Next, we supplement the ETH staking data with data from DeFi lending platforms, summarized in Table A2. Using data from the largest DeFi lending platform, AAVE, [Chaudhary, Kozhan, and Vishwanath-Natraj \(2023\)](#) document an average Ether lending rate of 0.129% with a maximum of 2.284% over the same sample. Similarly, they document an average USDC lending rate of 2.832% with a maximum of 8.957% over the same sample.

**Table A2**  
**Estimates for annual DeFi return,  $\psi$  (see Appendix A)**

| Data source                                   | Estimate for annual DeFi return $\psi$ | Application              |
|-----------------------------------------------|----------------------------------------|--------------------------|
| AAVE                                          | 1.66% average (2023-2024)              | ETH DeFi lending (AAVE)  |
| Academic source                               | Estimate for annual DeFi return $\psi$ | DeFi application         |
| Chaudhary, Kozhan, and Vishwanath-Natraj 2023 | 0.13% average (max 2.28%)              | ETH DeFi lending (AAVE)  |
| Chaudhary, Kozhan, and Vishwanath-Natraj 2023 | 2.832% average (max 8.957%)            | USDC DeFi lending (AAVE) |
| Park and Stinner 2023                         | 4.8% average                           | All assets (AAVE)        |
| Park and Stinner 2023                         | 1.2% average                           | All assets (Compound)    |

- Similarly, [Park and Stinner \(2023\)](#) demonstrate an average lending return of 4.8% across all pools on AAVE from December 2020 to February 2023 and 1.2% across all pools on Compound from May 2019 to February 2023.
- Finally, using data directly provided by AAVE (via <https://app.aave.com/markets/>), we can see that the average average ETH lending return was 1.66% over the same time period while the average USDC lending return over the last year (June 6, 2023, to June 6, 2024) is 5.39%.

## Appendix B: Proofs

### B.1 Proof of Lemma 1

**Proof.** The optimality of fees requires that for all  $c \leq c^P$ :

$$\phi^P(c) = \arg \max_{f: f \geq 0} \Pi^P - f - \frac{c}{\Lambda} \times (G(c^P) - G((\phi^P)^{-1}(f))) \quad (\text{B1})$$

with  $(\phi^P)^{-1}$  denoting the inverse function of  $\phi^P$  over  $(0, \phi(c^P))$ ,  $(\phi^P)^{-1}(f) \equiv c^P$  for  $f > \phi(c^P)$  and  $(\phi^P)^{-1}(0) \equiv 0$ . This generalized definition of the inverse function of  $\phi^P$  reflects that any user considering the out-of-equilibrium action of paying a fee higher than that paid in equilibrium internalizes that she would receive immediate service (i.e.,  $f > \phi(c^P)$  implies  $\{G(c^P) - G((\phi^P)^{-1}(f))\} = 0$ ). Moreover, any user paying a zero fee internalizes that she would have to wait for all other users before receiving service (i.e.,  $f = 0$  implies  $\{G(c^P) - G((\phi^P)^{-1}(f))\} = G(c^P)$ ).

The first-order condition for [Equation \(B1\)](#) is given by

$$-1 + \frac{c}{\Lambda} \cdot G'((\phi^P)^{-1}(f)) \cdot \frac{\partial}{\partial f} (\phi^P)^{-1}(f) = 0,$$

which after applying the inverse function theorem, imposing  $f_{(i,t)}^P = \phi^P(c_{(i,t)})$ , and rearranging yields

$$\frac{d\phi^P}{dc} = \frac{c}{\Lambda} G'(c) \quad (\text{B2})$$

This differential equation is defined over  $c_{(i,t)} \in [0, c^P]$  and has the boundary condition  $\phi^P(0) = 0$  (i.e., a zero fee is optimal for any agent with zero wait disutility). Accordingly, the unique equilibrium fee function,  $\phi^P$ , is given explicitly by:

$$\text{For all } (i, t): \phi^P(c) = \begin{cases} \frac{1}{\Lambda} \int_0^c x \, dG(x) & \text{if } c < c^P \\ 0 & \text{if } c \geq c^P \end{cases} \quad (\text{B3})$$

which gives the equilibrium realized fees [\(33\)](#). □

## B.2 Proof of Proposition 1

**Proof.** First note that the market value  $\mathcal{M}^{PoW}$  comes immediately from the market clearing condition (25) combined with the fee solution provided by (33). To determine the equilibrium return from mining, we start with Equation (22) and use the fact that

$$P_{t+2}^{PoW} B_{t+1} = P_{t+2}^{PoW} M_{t+1}(e^\rho - 1) = P_{t+2}^{PoW} M_{t+2} \frac{(e^\rho - 1)}{e^\rho} = \mathcal{M}^{PoW}(1 - e^{-\rho}),$$

which after substituting for  $\mathcal{M}^{PoW}$  from from (35) implies that

$$(B_{t+1} + \frac{\int f_{(i,t-1)}^{PoW} dG(c_{(i,t-1)})}{P_{t+1}^{PoW}}) P_{t+2}^{PoW} = (1 - e^{-2\rho})(1 - \omega^{PoW})G(c^{PoW}) + fees(c^{PoW}),$$

which after substituting into (22) and using the fact that  $\frac{P_{t+2}^{PoS}}{P_{t+1}^{PoS}} = \frac{\mathcal{M}^{PoW}}{\mathcal{M}^{PoW}} \cdot \frac{M_{t+1}}{M_{t+2}} = \frac{M_{t+1}}{M_{t+1}e^\rho} = e^{-\rho}$  yields our expression for  $R^{PoW}$ .

Next, we note that  $\omega^{PoW}$  is the value of  $\omega$  that equates the return from mining with the return from the DeFi investment. Namely, after substituting for  $H^*(1 + \kappa_h) = \omega^{PoW} G(c^{PoW})$  into (37) we can see that  $\omega^{PoW}$  is the solution to the following equation

$$R^{PoW} = \frac{\omega^{PoW} G(c^{PoW}) \cdot \frac{1-\delta}{1+\kappa_h} + (1 - e^{-2\rho})(1 - \omega^{PoW})G(c^{PoW}) + fees(c^{PoW})}{\omega^{PoW} G(c^{PoW})} = e^{-2\rho}(1 + \psi) = R^{DeFi}$$

which after rearranging and using the fact that  $\omega^{PoW} \leq 1$  yields our expression for  $\omega^{PoW}$ .

The remaining expressions are derived by substituting for  $\omega^{PoW}$ ,  $H^*(1 + \kappa_h)$ , and using the fact that in equilibrium  $\Pi_t^{PoW} = e^{-2\rho}(1 + \psi)$  because  $R^{PoW} = R^{DeFi} = e^{-2\rho}(1 + \psi)$  when  $\omega^{PoW} < 1$ . In contrast, when  $\omega^{PoW} = 1$  then  $\Pi_t^{PoW} = \frac{1-\delta}{1+\kappa_h} + \frac{fees(c^{PoW})}{G(c^{PoW})} \geq e^{-2\rho}(1 + \psi)$  with the inequality coming from the fact that whenever  $\frac{1-\delta}{1+\kappa_h} + \frac{fees(c^{PoW})}{G(c^{PoW})} < e^{-2\rho}(1 + \psi)$  then it must be the case that  $\omega^{PoW} < 1$ . Hence, in general  $\Pi_t^{PoW} = \Pi^{PoW} = \max\{e^{-2\rho}(1 + \psi), \frac{1-\delta}{1+\kappa_h} + \frac{fees(c^{PoW})}{G(c^{PoW})}\}$ . Finally, we note that the restriction  $\omega^{PoW} \leq 1$  implies that when

$$\frac{(1 - e^{-2\rho})G(c^{PoW}) + fees(c^{PoW})}{G(c^{PoW})(e^{-2\rho} \cdot \psi + 1 - \frac{1-\delta}{1+\kappa_h})} > 1$$

then  $\omega^{PoW} = 1$  and therefore  $H^*(1 + \kappa_h) = G(c^{PoW})$  while otherwise

$$H^*(1 + \kappa_h) = \frac{(1 - e^{-2\rho})G(c^{PoW}) + fees(c^{PoW})}{e^{-2\rho} \cdot \psi + 1 - \frac{1-\delta}{1+\kappa_h}}$$

hence,  $H^*(1 + \kappa_h) = \min\{\frac{(1 - e^{-2\rho})G(c^{PoW}) + fees(c^{PoW})}{e^{-2\rho} \cdot \psi + 1 - \frac{1-\delta}{1+\kappa_h}}, G(c^{PoW})\}$ .

Finally, to prove that the cutoff equilibrium is well-defined, we must show that  $c_{(i,t)} < c^{PoW}$  implies that  $U_{(i,t)}^{PoW} > 1 + \sigma$  and  $c_{(i,t)} > c^{PoW}$  implies that  $U_{(i,t)}^{PoW} \leq 1 + \sigma$ . To show this, first note that whenever  $\sup\{c \geq 0 : U^{PoW}(c, c) \geq 1 + \sigma\} > 0$  then there exists  $c^{PoW} > 0$  such that

$$U(c^{PoW}, c^{PoW}) = \pi^{PoW}(c^{PoW}) \times (\Pi^{PoW} - \frac{1}{\Lambda} \int_0^{c^{PoW}} x dG(x)) \geq 1 + \sigma$$

Further, for any  $c_{(i,t)} < c^{PoW}$  we note that

$$\frac{c_{i,t}}{\Lambda} (G(c^{PoW}) - G(c_{(i,t)})) = \frac{1}{\Lambda} \int_{c_{(i,t)}}^{c^{PoW}} c_{(i,t)} dG(x) < \frac{1}{\Lambda} \int_{c_{(i,t)}}^{c^{PoW}} x dG(x)$$

and therefore  $c_{(i,t)} < c^{PoW}$  implies

$$\begin{aligned} U_{(i,t)}^{PoW} &= \pi^{PoW}(c^{PoW}) \times (\Pi^{PoW} - \frac{1}{\Lambda} \int_0^{c_{(i,t)}} x dG(x) - \frac{c_{(i,t)}}{\Lambda} (G(c^{PoW}) - G(c_{(i,t)}))) > \\ \pi^{PoW}(c^{PoW}) \times (\Pi^{PoW} - \frac{1}{\Lambda} \int_0^{c_{(i,t)}} x dG(x) - \frac{1}{\Lambda} \int_{c_{(i,t)}}^{c^{PoW}} x dG(x)) &= U(c^{PoW}, c^{PoW}) = 1 + \sigma \end{aligned}$$

Further, if instead  $\sup\{c \geq 0 : U_{(i,t)}^{PoW}(c, c) \geq 1 + \sigma\} = 0$  then it must be the case that  $U_{(i,t)}^{PoW}(c, c) \leq 1 + \sigma$  for all  $c$  and therefore  $c^{PoW} = 0$ , in which case there is no Agent  $(i, t)$  such that  $c_{(i,t)} < c^{PoW} = 0$  coming from the fact that we have assumed  $c_{(i,t)} \geq 0$  for all  $(i, t)$ .

For the last step we must show that  $c_{(i,t)} > c^{PoW}$  implies that  $U_{(i,t)}^{PoW} \leq 1 + \sigma$ . Again we start with the case whereby  $\sup\{c \geq 0 : U_{(i,t)}^{PoW}(c, c) \geq 1 + \sigma\} > 0$  in which case  $c^{PoW} > 0$ . Then, we note that if any Agent  $(i, t)$  with wait disutility  $c_{(i,t)} > c^{PoW}$  were to switch from not adopting the blockchain to adopting the blockchain, they would receive immediate processing by offering a fee of

$f_{(i,t)} = \hat{f}(c^{PoW}) := \frac{1}{\Lambda} \int_0^{c^{PoW}} x dG(x)$ . Further, by offering a fee of  $f_{(i,t)} = \hat{f}(c^{PoW})$ , Agent  $(i, t)$  can guarantee  $U_{(i,t)}^{PoW} \big|_{f_{(i,t)} = \hat{f}(c^{PoW})} = U(c^{PoW}, c^{PoW}) = 1 + \sigma$ .

Next we note that if Agent  $(i, t)$  could increase their utility from adoption by offering a fee  $f_{(i,t)} \neq \hat{f}(c^{PoW})$  then this would contradict the fact that paying the fee of  $\hat{f}(c^{PoW})$  is optimal for the Agent  $(i', t)$  with  $c_{(i',t)} = c^{PoW}$ . In particular, if by offering any fee  $f_{(i,t)} < \hat{f}(c^{PoW})$ , Agent  $(i, t)$  could generate higher utility than  $1 + \sigma$ , then it must be the case that Agent  $(i', t)$  with  $c_{(i',t)} = c^{PoW} < c_{(i,t)}$  can obtain a utility strictly higher than  $1 + \sigma$  by offering the same fee  $f_{(i,t)} < \hat{f}(c^{PoW})$  as Agent  $(i', t)$  has a strictly lower wait disutility than Agent  $(i, t)$ , contradicting the optimality of the fees derived in Proposition 1. In contrast, it is clear that Agent  $(i, t)$  cannot be better off offering a fee  $f_{(i,t)} > \hat{f}(c^{PoW})$  as by doing so they obtain the same expected wait time (equal to zero) as when offering a fee of  $f_{(i,t)} = \hat{f}(c^{PoW})$ .

Finally, we note that if  $\sup\{c \geq 0 : U_{(i,t)}^{PoW}(c, c) \geq 1 + \sigma\} = 0$  then  $c^{PoW} = 0$  and therefore  $\pi^{PoW} = 0$  and thus  $U_{(i,t)}^{PoW} = 0 < 1 + \sigma$  for all Agents  $(i, t)$  with  $c_{(i,t)} > c^{PoW} = 0$ .  $\square$

### B.3 Proof of Proposition 2

**Proof.** Applying Equation (33) to Equation (25) (which is derived using the fact that  $\frac{P_t^P}{P_{t-1}^P} = \frac{\mathcal{M}_t^P}{\mathcal{M}_{t-1}^P} = e^{-\rho}$ ) yields the solution (44) for  $\mathcal{M}^{PoS}$ .

To derive the return  $R^{PoS}$ , we first note that  $S_{t+1} P_t^{PoS} = \mathcal{S}^*$ . Next, we use the fact that

$$P_{t+2}^{PoS} B_{t+1} = P_{t+2}^{PoS} M_{t+1} (e^\rho - 1) = P_{t+2}^{PoS} M_{t+2} \frac{(e^\rho - 1)}{e^\rho} = \mathcal{M}^{PoS} (1 - e^{-\rho}),$$

which after substituting for  $\mathcal{M}^{PoS}$  from from (35) implies that

$$(B_{t+1} + \frac{\int f_{(i,t-1)}^{PoS} dG(c_{(i,t-1)})}{P_{t+1}^{PoS}}) P_{t+2}^{PoS} = (1 - e^{-2\rho}) G(c^{PoS}) + fees(c^{PoS})$$

which after substituting into (23) and using the fact that  $\frac{P_{t+2}^{PoS}}{P_t^{PoS}} = e^{-2\rho}$  yields our expression for  $R^{PoS}$ .

Next, note that  $\omega^{PoS}$  is the value of  $\omega$  that equates the return from staking with the return from DeFi investment. Namely, using (46) and (27) we can see that  $\omega^{PoS}$  is derived as the solution to the following equation

$$R^{PoS} = e^{-2\rho} + \frac{(1 - e^{-2\rho})G(c^{PoS}) + fees(c^{PoS})}{\omega^{PoS} \cdot G(c^{PoS})} = e^{-2\rho}(1 + \psi) = R^{DeFi}$$

which after solving for  $\omega^{PoS}$  and using the fact that  $\omega^{PoS} \leq 1$  yields our expression.

We obtain the remaining expressions by substituting  $\omega^{PoS}$  into  $S^* = \omega^{PoS} G(c^{PoS})$ ,  $S^*$  into  $\pi^{PoS} = \min\{\frac{a}{\pi} \cdot S^*, 1\}$ , and using the fact that  $\Pi^{PoS} = e^{-2\rho}(1 + \psi)$ . Further, we note that the restriction  $\omega^{PoS} \leq 1$  implies that whenever

$$\frac{(1 - e^{-2\rho})G(c^{PoS}) + fees(c^{PoS})}{G(c^{PoS}) \cdot e^{-2\rho} \cdot \psi} > 1$$

then  $\omega^{PoS} = 1$  and therefore  $S^* = G(c^{PoW})$ . In contrast, whenever  $\omega^{PoS} < 1$  then

$$S^* = \frac{(1 - e^{-2\rho})G(c^{PoS}) + fees(c^{PoS})}{e^{-2\rho} \cdot \psi}$$

and therefore

$$S^* = \min\left\{\frac{(1 - e^{-2\rho})G(c^{PoS}) + fees(c^{PoS})}{e^{-2\rho} \cdot \psi}, G(c^{PoS})\right\}$$

Next, to derive our expression for  $U^{PoS}(c_{(i,t)}, c^{PoS})$ , we start with (5) and note that in the case in which  $\omega^{PoS} = 1$ , then  $S^* = G(c^{PoS})$  and therefore

$$\Pi^{PoS} = 1 + \frac{fees(c^{PoS})}{G(c^{PoS})}$$

Further,  $\omega^{PoS} = 1$  if and only if

$$1 + \frac{fees(c^{PoS})}{G(c^{PoS})} \geq e^{-2\rho}(1 + \psi)$$

and therefore

$$\Pi^{PoS} = \max\{e^{-2\rho}(1 + \psi), 1 + \frac{fees(c^{PoS})}{G(c^{PoS})}\}$$

Finally, to prove that the cutoff equilibrium is well-defined, we must show that  $c_{(i,t)} < c^{PoS}$  implies that  $U_{(i,t)}^{PoS} > 1 + \sigma$  and  $c_{(i,t)} > c^{PoS}$  implies that  $U_{(i,t)}^{PoS} \leq 1 + \sigma$ . To show this, first note that whenever  $\sup\{c \geq 0 : U^{PoS}(c, c) \geq 1 + \sigma\} > 0$  then there exists  $c^{PoS} > 0$  such that

$$U(c^{PoS}, c^{PoS}) = \pi^{PoS}(c^{PoS}) \times (\Pi^{PoS} - \frac{1}{\Lambda} \int_0^{c^{PoS}} x dG(x)) \geq 1 + \sigma$$

Further, for any  $c_{(i,t)} < c^{PoS}$  we note that

$$\frac{c_{(i,t)}}{\Lambda} (G(c^{PoS}) - G(c)) = \frac{1}{\Lambda} \int_{c_{(i,t)}}^{c^{PoS}} c_{(i,t)} dG(x) < \frac{1}{\Lambda} \int_{c_{(i,t)}}^{c^{PoS}} x dG(x)$$

and therefore  $c_{(i,t)} < c^{PoS}$  implies

$$U_{(i,t)}^{PoS} = \pi^{PoS}(c^{PoS}) \times (\Pi^{PoS} - \frac{1}{\Lambda} \int_0^{c_{(i,t)}} x dG(x) - \frac{c_{(i,t)}}{\Lambda} (G(c^{PoS}) - G(c_{(i,t)}))) >$$

$$\pi^{PoS}(c^{PoS}) \times (\Pi^{PoS} - \frac{1}{\Lambda} \int_0^{c_{(i,t)}} x dG(x) - \frac{1}{\Lambda} \int_0^{c^{PoS}} x dG(x)) = U(c^{PoS}, c^{PoS}) = 1 + \sigma$$

Further, if instead  $\sup\{c \geq 0 : U^{PoS}(c, c) \geq 1 + \sigma\} = 0$  then it must be the case that  $U_{(i,t)}^{PoS}(c, c) \leq 1 + \sigma$  for all  $c$  and therefore  $c^{PoS} = 0$ , in which case there is no Agent  $(i, t)$  such that  $c_{(i,t)} < c^{PoS} = 0$  coming from the fact that we have assumed  $c_{(i,t)} \geq 0$  for all  $(i, t)$ .

For the last step we must show that  $c_{(i,t)} > c^{PoS}$  implies that  $U_{(i,t)}^{PoS} \leq 1 + \sigma$ . Again we start with the case whereby  $\sup\{c \geq 0 : U^{PoS}(c, c) \geq 1 + \sigma\} > 0$  in which case  $c^{PoS} > 0$ . Then, we note that if any Agent  $(i, t)$  with wait disutility  $c_{(i,t)} > c^{PoS}$  were to switch from not adopting the blockchain to adopting the blockchain, they would receive immediate processing by offering a fee of  $f_{(i,t)} =$

$$\hat{f}(c^{PoS}) := \frac{1}{\Lambda} \int_0^{c^{PoS}} x dG(x). \text{ Further, by offering a fee of } f_{(i,t)} = \hat{f}(c^{PoS}), \text{ Agent } (i, t) \text{ can guarantee}$$

$$U_{(i,t)}^{PoS} |_{f_{(i,t)} = \hat{f}(c^{PoS})} = U(c^{PoS}, c^{PoS}) = 1 + \sigma.$$

Next we note that if Agent  $(i, t)$  could increase their utility from adoption by offering a fee  $f_{(i,t)} \neq \hat{f}(c^{PoS})$  then this would contradict the fact that paying the fee of  $\hat{f}(c^{PoS})$  is optimal for the Agent  $(i', t)$  with  $c_{(i',t)} = c^{PoS}$ . In particular, if by offering any fee  $f_{(i,t)} < \hat{f}(c^{PoS})$ , Agent  $(i, t)$  could generate higher utility than  $1 + \sigma$ , then it must be the case that Agent  $(i', t)$  with  $c_{(i',t)} = c^{PoS} < c_{(i,t)}$  can obtain a utility strictly higher than  $1 + \sigma$  by offering the same fee  $f_{(i,t)} < \hat{f}(c^{PoS})$  as Agent  $(i', t)$  has a strictly lower wait disutility than Agent  $(i, t)$ , contradicting the optimality of the fees derived in Proposition 1. In contrast, it is clear that Agent  $(i, t)$  cannot be better off offering a fee  $f_{(i,t)} > \hat{f}(c^{PoS})$  as by doing so they obtain the same expected wait time (equal to zero) as when offering a fee of  $f_{(i,t)} = \hat{f}(c^{PoS})$ .

Finally, we note that if  $\sup\{c \geq 0 : U^{PoS}(c, c) \geq 1 + \sigma\} = 0$  then  $c^{PoS} = 0$  and therefore  $\pi^{PoS} = 0$  and thus  $U_{(i,t)}^{PoS} = 0 < 1 + \sigma$  for all Agents  $(i, t)$  with  $c_{(i,t)} > c^{PoS} = 0$ .  $\square$

The following lemma presents an intermediary result that will be utilized in what follows.

**Lemma 2.** For any underlying parameters of the model:

1. Whenever  $c^{PoW} > 0$  then  $\hat{\pi}^{PoW}(c^{PoW}) < \frac{1}{\Gamma} \cdot G(c^{PoW})$ .
2. Whenever  $c^{PoS} > 0$  then  $\hat{\pi}^{PoS}(c^{PoS}) < \frac{\alpha}{\Gamma} \cdot G(c^{PoS})$ .

**Proof.** We will prove the result by contrapositive. In particular, first note that  $\hat{\pi}^{PoW}(c^{PoW}) = \frac{1}{\Gamma} \cdot G(c^{PoW})$  implies that  $H^*(1 + \kappa_h) = G(c^{PoW})$  and therefore  $\omega^{PoW} = 1$ . Similarly,  $\hat{\pi}^{PoS}(c^{PoS}) = \frac{\alpha}{\Gamma} \cdot G(c^{PoS})$  implies that  $S^* = G(c^{PoS})$  and therefore  $\omega^{PoS} = 1$ . Further, whenever  $\omega^{PoW} = 1$  and  $H^*(1 + \kappa_h) = G(c^{PoW})$  then

$$\Pi^{PoW} = \frac{1 - \delta}{1 + \kappa_h} + \frac{f_{ees}(c^{PoW})}{G(c^{PoW})}$$

Similarly, whenever  $\omega^{PoS} = 1$  and  $S^* = 1$  then  $\Pi^{PoS} = 1 + \frac{f_{ees}(c^{PoS})}{G(c^{PoS})}$ . Hence, using the fact that

$$f_{ees}(c^P) = \frac{1}{\Lambda} \int_0^{c^P} \int_0^c x dG(x) dG(c) \leq \int_0^{c^P} \int_0^{c^P} x dG(x) dG(c) = G(c^P) \cdot \frac{1}{\Lambda} \int_0^{c^P} x dG(x)$$

we can see that  $\frac{fees(c^P)}{G(c^P)} \leq \frac{1}{\Lambda} \int_0^{c^P} x dG(x)$ . Thus, whenever  $\omega^P=1$  then  $U^P(c^P, c^P) \leq 1 < 1+\sigma$  and therefore  $c^P=0$  for any  $p \in \{PoW, PoS\}$ .

#### B.4 Proof of Proposition 3

**Proof.** We will prove both results over two cases.

**Case 1:**  $c^{PoW}=c^{PoS}=+\infty$ : We will start by proving  $\alpha \leq \tilde{\alpha}$  implies  $\pi^{PoW} \geq \pi^{PoS}$  in this case. To do so, first note that  $\alpha \leq \tilde{\alpha}$  implies that

$$\frac{\alpha}{e^{-2\rho} \cdot \psi} \leq \frac{1}{e^{-2\rho} \cdot \psi + 1 - \frac{1-\delta}{1+\kappa_h}}$$

and therefore

$$\frac{\alpha}{\Gamma} \cdot \min\left\{\frac{(1-e^{-\rho})G(c)+fees(c)}{e^{-2\rho} \cdot \psi}, G(c)\right\} \leq \frac{1}{\Gamma} \cdot \min\left\{\frac{(1-e^{-\rho})G(c)+fees(c)}{e^{-2\rho} \cdot \psi + 1 - \frac{1-\delta}{1+\kappa_h}}, G(c)\right\}$$

for all  $c$  which implies that  $\hat{\pi}^{PoS}(c) < \hat{\pi}^{PoW}(c)$  for all  $c$ . Further, noting that  $\pi^{PoW} = \min\{\hat{\pi}^{PoW}(+\infty), 1\}$  and  $\pi^{PoS} = \min\{\hat{\pi}^{PoS}(+\infty), 1\}$  where both  $\hat{\pi}^{PoW}(+\infty)$  and  $\hat{\pi}^{PoS}(+\infty)$  are finite then implies that  $\pi^{PoW} \geq \pi^{PoS}$ .

Next we will prove, in this case whereby  $c^{PoW}=c^{PoS}=+\infty$ , that  $\pi^{PoW} > \pi^{PoS}$  implies  $\alpha < \tilde{\alpha}$ , which is the contrapositive of  $\alpha \geq \tilde{\alpha}$  implies  $\pi^{PoS} \geq \pi^{PoW}$ . To do so, note that  $\pi^{PoW} > \pi^{PoS}$  implies that  $\pi^{PoS} < 1$  and therefore:

$$\pi^{PoW} = \min\{\hat{\pi}^{PoW}(+\infty), 1\} > \hat{\pi}^{PoS}(+\infty) = \pi^{PoS}$$

which using the fact that  $G(+\infty)=1$  implies that  $\hat{\pi}^{PoW}(+\infty) = \frac{1}{\Gamma} \cdot \min\{H^*(1+\kappa_h), 1\} > \frac{\alpha}{\Gamma} \cdot \min\{S^*, 1\} = \hat{\pi}^{PoS}(+\infty)$ . Further, we know that  $c^{PoW}=c^{PoS}=+\infty$  implies by Lemma 2 that  $\hat{\pi}^{PoW}(+\infty) = \frac{1}{\Gamma} \cdot H^*(1+\kappa_h)$  and  $\hat{\pi}^{PoS}(+\infty) = \frac{\alpha}{\Gamma} \cdot S^*$ . Thus,  $\pi^{PoW} > \pi^{PoS}$  implies that  $\hat{\pi}^{PoW}(+\infty) > \hat{\pi}^{PoS}(+\infty)$  and therefore

$$\frac{1}{\Gamma} \cdot \frac{(1-e^{-2\rho})G(+\infty)+fees(+\infty)}{e^{-2\rho} \cdot \psi + 1 - \frac{1-\delta}{1+\kappa_h}} > \frac{\alpha}{\Gamma} \cdot \frac{(1-e^{-2\rho})G(+\infty)+fees(+\infty)}{e^{-2\rho} \cdot \psi}, \quad (B4)$$

which using the fact that  $(1-e^{-2\rho})G(+\infty)+fees(+\infty) > 0$  and finite, implies that whenever (B4) holds then, after rearranging, it must be the case that  $\alpha < \tilde{\alpha}$ .

**Case 2:**  $c^{PoW} < +\infty$  or  $c^{PoS} < +\infty$ : We will proceed without loss with the case whereby  $c^{PoW} < +\infty$ . In this case, we will first prove that  $\alpha \leq \tilde{\alpha}$  implies that  $\pi^{PoW} \geq \pi^{PoS}$ . Namely, whenever  $\alpha \leq \tilde{\alpha}$  then it must be the case that  $\hat{\pi}^{PoW}(c) \geq \hat{\pi}^{PoS}(c)$  for all  $c$ , which implies that  $\pi^{PoW}(c^{PoS}) \geq \pi^{PoS}(c^{PoS})$  and therefore

$$1+\sigma \leq \pi^{PoS}(c^{PoS})(\Pi^{PoS}(c^{PoS}) - \frac{1}{\Lambda} \int_0^{c^{PoS}} x dG(x)) \leq \pi^{PoW}(c^{PoS})(\Pi^{PoS}(c^{PoS}) - \frac{1}{\Lambda} \int_0^{c^{PoS}} x dG(x)) \quad (B5)$$

We will now show that this condition implies that  $c^{PoS} < +\infty$ . Namely, if  $c^{PoS} = +\infty$  then it must be the case that

$$\begin{aligned} 1+\sigma &\leq \min\{\hat{\pi}^{PoS}(+\infty), 1\}(\Pi^{PoS}(+\infty) - \frac{1}{\Lambda} \int_0^{+\infty} x dG(x)) \\ &\leq \min\{\hat{\pi}^{PoW}(+\infty), 1\}(\Pi^{PoS}(+\infty) - \frac{1}{\Lambda} \int_0^{+\infty} x dG(x)) \end{aligned}$$

and therefore, using the fact that  $\Pi^{PoS}(c) = \Pi^{PoW}(c)$  for all  $c$ , then it must be the case that  $c^{PoW} = +\infty$ , contradicting the assumption that  $c^{PoW} < +\infty$ .

Finally, when  $c^{PoS} < +\infty$  and (B5) holds then, again using the fact that  $\Pi^{PoS}(c) = \Pi^{PoW}(c)$  for all  $c$ , it must be the case that

$$\begin{aligned} 1 + \sigma &= \pi^{PoS}(c^{PoS})(\Pi^{PoS}(c^{PoS}) - \frac{1}{\Lambda} \int_0^{c^{PoS}} x dG(x)) \\ &\leq \pi^{PoW}(c^{PoS})(\Pi^{PoW}(c^{PoS}) - \frac{1}{\Lambda} \int_0^{c^{PoS}} x dG(x)), \end{aligned}$$

Further, once this condition holds, then there must exist  $c' \geq c^{PoS}$  such that

$$1 + \sigma \leq \min\{\hat{\pi}^{PoW}(c'), 1\}(\Pi^{PoW}(c') - \frac{1}{\Lambda} \int_0^{c'} x dG(x)),$$

which implies that  $c^{PoW} \geq c^{PoS}$  and therefore  $\pi^{PoW}(c^{PoW}) \geq \pi^{PoW}(c^{PoS}) \geq \pi^{PoS}(c^{PoS})$ .

We will now proceed to prove that  $\alpha \geq \tilde{\alpha}$  implies  $\pi^{PoS} \geq \pi^{PoW}$  by proving the contrapositive  $\pi^{PoW} > \pi^{PoS}$  implies  $\alpha < \tilde{\alpha}$ . We again proceed without loss with the case whereby  $c^{PoW} < +\infty$ . In that case, we know that  $\pi^{PoW} > \pi^{PoS}$  implies  $\pi^{PoW} > 0$  and therefore  $c^{PoW} > 0$ . Thus,  $c^{PoW} > 0$  and  $c^{PoW} < +\infty$  implies that  $c^{PoW}$  is the maximal solution to

$$1 + \sigma = \pi^{PoW}(c^{PoW})(\Pi^{PoW}(c^{PoW}) - \frac{1}{\Lambda} \int_0^{c^{PoW}} x dG(x))$$

We start by considering the case whereby  $c^{PoS} > 0$ . First, note that by Lemma 2 we know that  $c^{PoW} > 0$  implies  $\pi^{PoW}(c^{PoW}) = \min\{\frac{1}{\Gamma} \cdot H^*(1 + \kappa_h), 1\}$  and  $\Pi^{PoW}(c^{PoW}) = \max\{e^{-2\rho}(1 + \psi), 1 + \frac{fees(c^{PoW})}{G(c^{PoW})}\} = e^{-2\rho}(1 + \psi)$ . Further, if  $c^{PoS} > 0$  then again by Lemma 2 we know that  $\pi^{PoS}(c^{PoS}) = \min\{\frac{\alpha}{\Gamma} \cdot S^*, 1\}$ , and  $\Pi^{PoS}(c^{PoS}) = \max\{e^{-2\rho}(1 + \psi), 1 + \frac{fees(c^{PoS})}{G(c^{PoS})}\} = e^{-2\rho}(1 + \psi)$ . In addition,  $\pi^{PoW} > \pi^{PoS}$  implies that  $\pi^{PoS} < 1$  and therefore  $\pi^{PoS} = \frac{\alpha}{\Gamma} \cdot S^*$ .

Next note that in this case,  $c^{PoS} > 0$  and  $\pi^{PoW}(c^{PoW}) > \pi^{PoS}(c^{PoS})$  implies that

$$1 + \sigma = \pi^{PoW}(c^{PoW})(e^{-2\rho}(1 + \psi) - \frac{1}{\Lambda} \int_0^{c^{PoW}} x dG(x)) > \pi^{PoS}(c^{PoS})(e^{-2\rho}(1 + \psi) - \frac{1}{\Lambda} \int_0^{c^{PoW}} x dG(x))$$

and therefore it must be the case that  $c^{PoS} < c^{PoW}$ . Further, whenever  $c^{PoS} < c^{PoW}$  then it must be the case that

$$1 + \sigma = \pi^{PoW}(c^{PoW})(e^{-2\rho}(1 + \psi) - \frac{1}{\Lambda} \int_0^{c^{PoW}} x dG(x)) > \pi^{PoS}(c^{PoW})(e^{-2\rho}(1 + \psi) - \frac{1}{\Lambda} \int_0^{c^{PoW}} x dG(x))$$

and therefore  $\pi^{PoW}(c^{PoW}) > \pi^{PoS}(c^{PoW})$ . Finally, we note that  $\pi^{PoW}(c^{PoW}) > \pi^{PoS}(c^{PoW})$  when  $\pi^{PoS} < 1$  implies that

$$\min\left\{\frac{1}{\Gamma} \cdot \frac{(1 - e^{-2\rho})G(c^{PoW}) + fees(c^{PoW})}{e^{-2\rho} \cdot \psi + 1 - \frac{1-\delta}{1+\kappa_h}}, 1\right\} > \frac{\alpha}{\Gamma} \cdot \frac{(1 - e^{-2\rho})G(c^{PoW}) + fees(c^{PoW})}{e^{-2\rho} \cdot \psi}$$

and therefore after rearranging it must be the case that  $\alpha < \tilde{\alpha}$ .

Finally, we return to the case whereby  $c^{PoS}=0$ . First, note that if  $c^{PoS}=0$  and  $\alpha \geq \tilde{\alpha}$  then it must be the case that  $c^{PoW}=0$  contradicting the fact that  $\pi^{PoW} > \pi^{PoS} \geq 0$ . Namely, suppose by contradiction that  $c^{PoS}=0$ ,  $\pi^{PoW} > \pi^{PoS}$  but  $\alpha \geq \tilde{\alpha}$ . In that case,  $\pi^{PoW} > \pi^{PoS}$  implies  $c^{PoW} > 0$  and therefore recalling that we are in the case whereby  $c^{PoW} < +\infty$  then  $c^{PoW}$  is the maximal solution to

$$1 + \sigma = \pi^{PoW}(c^{PoW})(e^{-2\rho}(1 + \psi) - \frac{1}{\Lambda} \int_0^{c^{PoW}} x dG(x))$$

Further,  $\alpha \geq \tilde{\alpha}$  implies that  $\frac{\alpha}{\Gamma} S^*(c) \geq \frac{1}{\Gamma} H^*(c)(1 + \kappa_h)$  for all  $c$  and therefore using the fact that  $S^*(c) = \omega^{PoS} \cdot G(c) \leq G(c)$  implies that

$$\hat{\pi}^{PoS}(c) = \min\left\{\frac{\alpha}{\Gamma} S^*(c), \frac{\alpha}{\Gamma} G(c)\right\} \geq \frac{1}{\Gamma} H^*(c)(1 + \kappa_h) = \hat{\pi}^{PoW}(c)$$

for all  $c$  whenever  $\alpha \geq \tilde{\alpha}$ , where the last equality comes from Lemma 2 and the fact that  $c^{PoW} > 0$ . Thus, we know that whenever  $\alpha \geq \tilde{\alpha}$  then  $\pi^{PoS}(c) = \min\{\hat{\pi}^{PoS}(c), 1\} \geq \min\{\hat{\pi}^{PoW}(c), 1\} = \pi^{PoW}(c)$ . Yet, whenever this is the case, then we know that

$$1 + \sigma = \pi^{PoW}(c^{PoW})(e^{-2\rho}(1 + \psi) - \frac{1}{\Lambda} \int_0^{c^{PoW}} x dG(x)) \leq \pi^{PoS}(c^{PoW})(e^{-2\rho}(1 + \psi) - \frac{1}{\Lambda} \int_0^{c^{PoW}} x dG(x)) \leq$$

$$\pi^{PoS}(c^{PoW})(\max\{e^{-2\rho}(1 + \psi), 1 + \frac{f_{ees}(c^{PoW})}{G(c^{PoW})}\} - \frac{1}{\Lambda} \int_0^{c^{PoW}} x dG(x))$$

and therefore it must be the case that  $0 = c^{PoS} \geq c^{PoW}$  and therefore  $c^{PoW} = 0$ , contradicting the fact that  $\pi^{PoW} > \pi^{PoS}$ .  $\square$

### B.5 Proof of Proposition 4

**Proof.** First suppose that the investor time horizon is  $T$  and suppose that (54) holds. Then, note that

$$\frac{\log(1 + \delta_T)}{\log(1 + \psi_T)} = \frac{\log(1 + \delta_{1Y})}{\log(1 + \psi_{1Y})} = \lim_{T \rightarrow 0^+} \frac{(1 + \delta_{1Y})^T - 1}{(1 + \psi_{1Y})^T - 1} \leq \frac{(1 + \delta_{1Y})^T - 1}{(1 + \psi_{1Y})^T - 1} = \frac{\delta_T}{\psi_T}$$

Therefore, when (54) holds then using the above inequality we can see that

$$\lambda \leq \frac{1}{2} \left(1 + \frac{\log(1 + \delta_T)}{\log(1 + \psi_T)}\right) \leq \frac{1}{2} \left(1 + \frac{\delta_T}{\psi_T}\right)$$

Finally,  $\eta \leq \frac{2}{3}$  combined with  $\lambda \leq \frac{1}{2} \left(1 + \frac{\delta_T}{\psi_T}\right)$  implies that

$$\alpha = \frac{1 - \eta}{\eta \cdot \lambda} \geq \frac{1}{2\lambda} \geq \frac{\psi_T}{\psi_T + \delta_T} \geq \frac{e^{-2\rho} \cdot \psi_T}{e^{-2\rho} \cdot \psi_T + 1 - \frac{1 - \delta_T}{1 + \kappa_h}} = \zeta$$

$\square$

### B.6 Proof of Corollary 1

To prove this result, note that Tables 1 and 2 demonstrate a parameter range of  $\psi_{1Y} \in [3.73\%, 4.85\%]$  and  $\delta_{1Y} \in [46.5\%, 89\%]$ . Thus, noting that  $\delta_{1Y} > \psi_{1Y}$  for all  $\psi_{1Y} \in [3.73\%, 4.85\%]$

and all  $\delta_{1Y} \in [46.5\%, 89\%]$ , then the result of Proposition 4 holds. Therefore, a PoS blockchain generates higher equilibrium security than a PoW blockchain whenever

$$\lambda \leq \Upsilon(\psi_{1Y}, \delta_{1Y}) := \frac{1}{2} \times (1 + \frac{\log(1 + \delta_{1Y})}{\log(1 + \psi_{1Y})})$$

Thus, noting that  $\Upsilon(\psi_{1Y}, \delta_{1Y})$  is decreasing in  $\psi_{1Y}$  and increasing in  $\delta_{1Y}$  then implies that  $\Upsilon(\psi_{1Y}, \delta_{1Y})$  achieves its minimum over the range  $\psi_{1Y} \in [3.73\%, 4.85\%]$  and  $\delta_{1Y} \in [46.5\%, 89\%]$  when  $\psi_{1Y} = 4.85\%$  and  $\delta_{1Y} = 46.5\%$ . Thus, noting that  $\Upsilon(4.85\%, 46.5\%) = 453\%$  implies that under the most conservative estimates (i.e.,  $\psi_{1Y} = 4.85\%$  and  $\delta_{1Y} = 46.5\%$ ) PoS blockchains are more secure than PoW blockchains whenever  $\lambda \leq 453\%$ .

Noting again that  $\Upsilon(\psi_{1Y}, \delta_{1Y})$  is decreasing in  $\psi_{1Y}$  and increasing in  $\delta_{1Y}$  then implies that  $\Upsilon(\psi_{1Y}, \delta_{1Y})$  achieves its minimum over the range  $\psi_{1Y} \in [3.73\%, 4.32\%]$  and  $\delta_{1Y} \in [55\%, 89\%]$  when  $\psi_{1Y} = 4.32\%$  and  $\delta_{1Y} = 55\%$  so that under these less conservative estimates (i.e.,  $\psi_{1Y} = 4.32\%$  and  $\delta_{1Y} = 55\%$ ) PoS blockchains are more secure than PoW blockchains whenever  $\lambda \leq \Upsilon(4.32\%, 55\%) = 568\%$ .

### B.7 Proof of Proposition 5

**Proof.** Fix  $\bar{\Gamma}' \in \Sigma_0^{PoS}$ . Then, by the definition of  $\Sigma_0^{PoS}$ , when the attacker's maximal benefit is  $\bar{\Gamma}'$ , there exists  $\underline{\Lambda}^{PoS}$  such that for all  $\Lambda > \underline{\Lambda}^{PoS}$  it is the case that  $\pi^{PoS} = 0$ . Yet, whenever  $\alpha \geq \zeta$  then  $\pi^{PoW} \leq \pi^{PoS}$  for all  $\Lambda$  and any maximal attacker benefit  $\bar{\Gamma}$ . Therefore whenever the attacker's benefit is  $\bar{\Gamma}'$  it must be the case that  $\pi^{PoW} = 0$  for all  $\Lambda > \underline{\Lambda}^{PoS}$ . Hence, it must be the case that  $\bar{\Gamma}' \in \Sigma_0^{PoW}$ .  $\square$

### B.8 Proof of Proposition 6

**Proof.** Fix the attacker's maximal benefit at  $\bar{\Gamma} > 0$ . By the definition of  $\bar{\Lambda}_0^{PoW}$  we know that for all  $\epsilon > 0$  there exists  $\Lambda \in (\bar{\Lambda}_0^{PoW} - \epsilon, \bar{\Lambda}_0^{PoW})$  such that  $\pi^{PoW} > 0$  whenever the blockchain scale is  $\Lambda$ . Further, when  $\alpha \geq \zeta$  this implies that  $\pi^{PoS} \geq \pi^{PoW} > 0$  when the blockchain scale is  $\Lambda$ . Yet, the fact that there exists  $\Lambda \in (\bar{\Lambda}_0^{PoW} - \epsilon, \bar{\Lambda}_0^{PoW})$  such that  $\pi^{PoW} > 0$  when the blockchain scale is  $\Lambda$  for all  $\epsilon > 0$  implies that for all  $\epsilon > 0$  there exists  $\Lambda \in (\bar{\Lambda}_0^{PoW} - \epsilon, \bar{\Lambda}_0^{PoW})$  such that  $\pi^{PoS} > 0$  whenever the blockchain scale is  $\Lambda$  and therefore  $\bar{\Lambda}_0^{PoS} \geq \bar{\Lambda}_0^{PoW}$ .  $\square$

### B.9 Proof of Proposition 7

**Proof.**  $\Sigma_1^{PoW} \subseteq \Sigma_1^{PoS}$ . Fix  $\bar{\Gamma}' \in \Sigma_1^{PoW}$ . Then by the definition of  $\Sigma_1^{PoW}$  it must be the case that when the attacker's benefit is  $\bar{\Gamma}'$  then there exists  $\underline{\Lambda}^{PoW}$  such that  $\Lambda > \underline{\Lambda}^{PoW}$  implies that  $\pi^{PoW} = 1$ . Further, whenever  $\alpha \geq \zeta$  then we know that  $\pi^{PoS} \geq \pi^{PoW}$  for all  $\Lambda$  and all levels of the attacker's maximal benefit  $\bar{\Gamma}$ . Hence, whenever the attacker's maximal benefit is  $\bar{\Gamma}'$  then  $\pi^{PoS} \geq \pi^{PoW} = 1$  for all  $\Lambda > \underline{\Lambda}^{PoW}$  and therefore  $\bar{\Gamma}' \in \Sigma_1^{PoS}$ .  $\square$

### B.10 Proof of Proposition 8

**Proof.** Fix the attacker's maximal benefit at  $\bar{\Gamma} > 0$  and note that if  $\bar{\Lambda}_1^{PoW} = +\infty$  then the result follows trivially. Otherwise,  $\bar{\Lambda}_1^{PoW} < +\infty$  and, in this case, we know that whenever  $\Lambda > \bar{\Lambda}_1^{PoW}$  then it must be the case that  $\pi^{PoW} = 1$ . Further, when  $\alpha \geq \zeta$  this implies that  $\pi^{PoS} \geq \pi^{PoW} = 1$  and therefore  $\pi^{PoS} = 1$  whenever  $\Lambda > \bar{\Lambda}_1^{PoW}$ . Hence, it must be the case that  $\bar{\Lambda}_1^{PoS} \leq \bar{\Lambda}_1^{PoW}$ .  $\square$

**Code Availability:** The replication code is available in the Harvard Dataverse at <https://doi.org/10.7910/DVN/5VTIMI>.

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